Investment Tax Credits and the Response of Firms*

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Abstract

This paper estimates separately direct and spillover effects of investment tax credits on input choices of firms by exploiting a shift in the tax credit rate of small compared to large manufacturing firms in Germany in 1999. I find that tax credits lead to a sizable increase in both investment and the number of employees of firms. While there is little evidence that the average firm adjusts its skill mix or occupation structure, firms in industries with a larger share of information and communication technology (ICT) are more likely to shift towards high-skilled labor and occupations. The employment effect materializes through the hiring of additional employees rather than a decrease in separations and the unemployed constitute about half of these hires. Local spillovers between firms lead to an additional positive employment effect and are therefore important when considering the cost effectiveness of tax credits. The effect tends to be stronger between firms that are in the same industry and is close to zero for the service sector.

JEL Classification: H25, H32, J23, L53, O25, R11
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Governments have long used tax policy with the intention to improve economic outcomes in times of weakness. Investment tax incentives such as investment tax credits and special depreciation allowances are particularly common, since altering firm investment is thought to be key for business cycle fluctuations and long term growth (Goolsbee, 1998).[1] Starting with Hall and Jorgenson (1967) a large literature has emerged on the effect of tax incentives on firm investment. Empirical research surveyed in Hassett and Hubbard (2002) and subsequent studies suggest that firms indeed respond by adjusting investment. However, directly incentivizing investments can have distributional consequences in the allocation of resources between industries, firms and workers (Chirinko, 2000).

To evaluate tax policies comprehensively, it is therefore important to understand how increases in investment among targeted firms may propagate to other economic agents. The policy implications in a scenario where the benefits mostly stay with capital owners of a selected group of firms are different from a scenario where investment tax incentives trigger the expansion of production, lead to higher labor demand and create spillover effects throughout an economy. Given that such an analysis requires unusually rich micro-level data and at the same time an estimation strategy that can identify direct and spillover effects separately, there is little known about such behavior.

In this paper, I use a tax policy in Germany to investigate the effect of investment tax credits on firm investment, employment and employment composition, and I quantify the relevance of spillover effects between firms. The tax policy was introduced in 1991, directly after German reunification, to counter the considerable economic differences that had developed during separation. East Germany had roughly half the GDP per capita and a much higher unemployment rate of 13%. The program provided investment tax credits to firms to reduce the costs for investment. Firms could recover up to 27.5% of investment costs, a substantial financial relief which led to additional government expenses of €1–2 billion per year.

I exploit variation in the tax credit rate with respect to firm size and time. During the period of analysis, manufacturing firms with at most 250 employees at the beginning of a business year were eligible for a tax credit rate of 10% whereas for those with more than 250 employees the rate was 5%. In 1999 a change in the rates amplified this differential treatment in favor of firms below the cutoff, adjusting the tax credit rate to 20% for firms below the cutoff and 10% for firms above the cutoff. This led to a relative decrease in capital costs for smaller firms.

For the empirical analysis, I use two distinct administrative datasets that collect detailed panel information on investment and employment for a large share of German

[1] Important policy examples for the U.S. are the use of investment tax credits in the Revenue Act of 1962, and the use of special depreciation allowances in the Job Creation and Worker Assistance Act of 2002 and the Economic Stimulus Act of 2008.

firms.[2] I implement a difference-in-differences estimation comparing the behavior of firms below and above the cutoff over the period 1995–2004. I include industry-year and labor market-year fixed effects to control for time-varying industry shocks and differential regional development, and I exclude observations close to the cutoff to avoid interference from bunching behavior. In a second step, I compare how firms react to the policy change depending on the share of firms below the cutoff within a labor market to estimate spillover effects.

The estimation strategy is guided by a theoretical firm production framework. In the model an increase in investment tax credits leads to an unambiguous positive effect on investment. But the effect on employment and employment composition is uncertain and depends on the level of substitution between capital and different labor types. Further, I allow that spillovers influence firm productivity within local labor markets which may generate an additional impact on investment and employment decisions. The spillover effects are larger within labor markets where there is a higher share of firms that receive tax credits. Thus, firm-level and regional variation allow the separate estimation of direct firm adjustment effects and spillover effects.

My first empirical finding is that tax credits have a substantial effect on investment. The increase in the relative tax credit rate leads to 23.4 log points higher overall investment and 25.1 log points higher equipment investment at the intensive margin. Given the underlying relative change in capital costs of 8.2%, the overall investment response corresponds to an elasticity with respect to capital costs of 2.8. This result is larger than findings summarized in Hassett and Hubbard (2002) that range of 0.5 to 1.0 but smaller than the elasticity of 7.2 by Zwick and Mahon (2017). Looking at the dynamic effects, the intensive margin investment response picks up immediately after the policy change. The effects gets stronger in the subsequent year and stabilizes at a higher level thereafter. There is however no discernible effect on the probability of investing.

I then consider the effect of the policy change on labor inputs and find a positive effect of tax credits on overall employment. Based on the preferred estimate, the change in the tax credit rate leads to 8.7 log points higher employment. This effect is similar when considering full-time workers, workers in regular employment or full-time workers in regular employment. Most of the effect materializes in the year after the policy change and there is a slight further increase in subsequent years. These adjustments in capital and labor inputs translate into more output, measured in terms of revenue. Estimates for the revenue effect however tend to be more volatile and significance levels are lower. The response to the tax credit rate change for domestic revenue amounts to 8.3 log points.

[2] I use the term "firm" interchangeably with "establishment" throughout. The main analysis is conducted at establishment level due to the level of aggregation in the datasets. The policy cutoff, however, is at firm level. I will show evidence that this simplification does not influence the qualitative results by considering single-establishment firms in a robustness test.

Taking these results together, tax credits are an effective tool to induce higher investment and these adjustments lead to more employment in firms. Further, I study different channels through which firms adjust their labor inputs. I find that the increase in employment is almost exclusively driven by hiring additional employees rather than fewer separations. There is actually a relative increase in separations for the treated firms, although the estimates are small and statistically insignificant. Among the additional hires, a share of 49% was unemployed one year prior which is similar to the share among all hires.

The employment effect might mask important heterogeneity by skill. Considering capital-skill complementarity, if new equipment substitutes for low-skilled tasks but requires high-skilled labor for its operation, the reduction in investment costs would lead to a shift towards highly educated labor and high-skilled occupations. I consider the ratios of various skill levels: college educated versus non-college educated employees (all or those with full-time employment), and high-skilled occupations including engineers and managers versus low- and medium-skilled occupations such as machine operators. All point estimates are close to zero and statistically insignificant. Given the magnitude of the standard errors, it is possible to reject estimates of all but modest skill composition changes. Thus, the added capital in firms does not shift employment opportunities towards high-skilled labor or occupations. I further consider the possibility that the shift happened only among manual labor by examining the effect on the ratio of medium-skilled manual occupations to low-skilled manual occupations, and between manual and service occupations. There is a zero effect for these ratios as well. The literature on technological change considers information and communication technology (ICT) as strongly complementary with high-skilled labor. When analyzing heterogeneous effects across industries, I find that industries with a higher share of ICT in investment and capital are more likely to shift towards college educated employees and high-skilled occupations.

To verify that these results are not driven by the research design or its implementation, I perform various robustness checks. First, plotting the raw data year by year and checking pre-treatment year estimates in the dynamic specifications, I find no differential pre-treatment behavior between treatment and control group which supports a common trends assumption. Second, adjustments in inputs are apparent directly in the year after the policy change providing a direct link between policy and firm behavior. Third, the results are robust to different sample selection procedures. Average effects are stable to selecting narrower or wider firm size intervals and to excluding different firm size intervals around the cutoff. Finally, in a placebo test where I select only treated firms and assume various policy irrelevant firm size cutoffs, I obtain insignificant estimates.

The results so far all concern the direct effect of tax credits but they leave out the potential further contribution from spillovers. When exploiting variation at the level of

the labor market, I find that the employment effect is larger for firms in labor markets with a larger share of firms below the cutoff. This result is in line with the theoretical model and implies that spillovers within a labor market lead to on average 10.7 log points higher firm employment among treated and control firms. I find that this effect tends to be larger for firms within the same industry and I do not find any significant effect on the service industries. These results suggest that similarities between firms are important which could be the case because of input-output linkages or knowledge spillovers. Local multipliers seem to be less important because a more prosperous labor market would generate higher demand for all types of goods including services. When quantifying the costs and benefits of the policy, I find that government expenses for investment tax credits of on average €11,346 a year are sufficient for the employment of one additional worker. When also considering spillover effects, the same government expenses create another employed worker. Spillovers are therefore important when considering the cost effectiveness of investment tax credits.

This paper contributes to a long literature that quantifies the effect of capital cost changes on firm investment. Earlier studies heavily rely on a structural approach and find surprisingly small responses in investment (Abel, 1980; Summers, 1981; Abel and Blanchard, 1986).[3] Strong assumptions about the capital formation process and limited variation in capital costs may be the reason for these findings since the same estimation approach using cross-sectional variation due to tax policy elicits larger effects (Cummins et al., 1994; Edgerton, 2010). The recent literature predominantly shifts towards the analysis of specific tax incentive programs that introduce cross-sectional variation (House and Shapiro, 2008; Yagan, 2015; Maffini et al., 2016; Zwick and Mahon, 2017) and finds even larger responses in investment behavior. These recent studies also find evidence that there are important heterogeneous effects in the investment response.

I study firm investment in a novel setting, using cross-sectional variation in the tax credit program in Germany. The particular setup of the program with changes in tax credit rate with respect to firm size allows to comprehensively analyze the impact of capital costs on a variety of firm outcomes and to study the underlying adjustment mechanisms. Foremost, the setting creates variation at firm and labor market level to quantify direct effects and spillover effects separately and it provides a way for directly controlling for industry-specific shocks and differences in regional development. Since the number of employees is available in many firm-level datasets, it is also possible to expand the analysis beyond investment to employment, employment composition and revenue. The program regulations and the background of introduction are otherwise comparable to programs in countries such as the U.S. and results are therefore likely meaningful in more general settings.

[3] The elasticities are considered *too* small, since they imply extremely high capital adjustment costs.

In my setting, I provide evidence that there is a sizable investment response to a capital costs reduction. The effect size is in line with findings by Maffini et al. (2016) and Zwick and Mahon (2017), but it is considerably larger to findings surveyed in Hassett and Hubbard (2002). Given the positive investment response, I then focus on employment outcomes and the influence of spillovers to understand how this response propagates through the economy. To the best of my knowledge, I am the first to comprehensively evaluate such behavior in the context of investment tax incentives. I can thereby assess the concern that such incentives have distributional consequences. I find a positive effect on employment, no effect on employment composition and important spillover effects which suggests that the incentives propagate through the economy. I conclude that distributional consequences are limited. Suárez Serrato and Zidar (2016) and Fuest et al. (2018) assess distributional effects on workers and firm owners for the case of corporate taxes and use regional variation for estimation. They do not consider spillover effects separately.

My analysis of spillover effects relates to Greenstone et al. (2010) and Gathmann et al. (2017). They exploit exogenous variation in firm openings and closings respectively to study spillover effects within regions.[4] Gathmann et al. (2017) find that spillovers amplify the effect by 35% which is similar to my result of 49%. Equal to their paper, I consider the importance of agglomeration spillovers compared to local multipliers. I find similar results, in that local multipliers do not influence the service sector but there is evidence that firms in the same industry have larger spillover effects. This is in contrast to Moretti and Thulin (2013) that report sizable local multiplier effects. The spillover analysis also provides a link between the existing firm-level studies and the macro-level time series studies such as Blanchard and Perotti (2002), Romer and Romer (2010) and Mertens and Ravn (2013) since spillovers are so far not considered on a firm-level but are implicitly included on a macro-level

Furthermore, my paper speaks to the large literature on place-based policies and their effects on regions (Becker et al., 2010, 2013; Busso et al., 2013; Kline and Moretti, 2014a; Criscuolo et al., 2016) and firms (Bronzini and de Blasio, 2006; Cerqua and Pellegrini, 2014). In these studies there is a stronger focus on employment, but the evidence on the employment effect is inconclusive. This may be because the analyzed policies often combine a mix of regional and firm specific incentives, including reductions in capital costs. I add to this literature by studying one particular incentive, investment tax incentives, and study adjustments in input and output independent of other influences.

Finally, by considering heterogeneous effects across labor types, my findings also speak to the often voiced concern in this literature that such policies can lead to unwanted redistribution in welfare (Glaeser and Gottlieb, 2008; Kline and Moretti, 2014b; Neumark

[4]The overall literature on spillovers is much larger. For an overview particularly concerning agglomeration economies see Combes and Gobillon (2015).

and Simpson, 2015). The literature on the shift of production technology towards automation (e.g. Acemoglu and Autor, 2011) and capital in general (Krusell et al., 2000) suggests that there can be an advantage for high-skilled over low-skilled labor. The tax credit program does not have adverse effects on the skill composition on average. It is also beneficial for previously unemployed individuals. However, I find an influence of ICT on firms shifting towards high-skilled labor. Similar to Akerman et al. (2015), this result points to potential adverse effects of government programs supporting ICT investments.

The structure of the paper is as follows. Section 1 explains the policy intervention in more detail focusing on the specific regulations that are important for the empirical analysis. Section 2 introduces a theoretical framework that provides intuition for expected firm behavior with heterogeneous labor types and spillovers. Section 3 explains the estimation strategy. Section 4 provides details on the data including descriptive statistics and sample selection. Section 5 presents the main results and Section 6 relates the results in a back-of-the-envelope cost-benefit analysis. Section 7 concludes.

1 Policy Intervention

After the second world war, Germany split into two countries, West Germany and East Germany.[5] While West Germany experienced continued growth with a market-based economy, East Germany faced large war reparations and inefficiencies in its communist economic system. The fall of communism throughout Eastern Europe led to the reunification of Germany in 1990. The diverging prior development however created a country with economically disparate regions. Over the period 1991 to 1994, East Germany had on average 46% lower GDP per capita, 47% lower capital per worker, 30% lower earnings per worker and an unemployment rate of 13.4% compared to 7.1% in West Germany (see Figure 1). To speed up economic convergence, the government provided considerable financial support to regions in East Germany. Besides cash transfers to private households and large infrastructure investments, efforts were focused on increasing the capital stock of firms. The most salient policy in this respect was an investment tax credit program (*Investitionszulagengesetz*) which is at the center of this paper. [6]

The program started immediately after reunification in 1991 and lasted until 2013. It provided tax credits for equipment investments to firms located in East Germany and West Berlin. From 1999, it also covered investments in structures. At the beginning of the program, firms of all industries were eligible for the program but over time access became more restricted and by 1997 coverage applied almost exclusively to manufacturing

[5] During the time of separation the official designation for West Germany was *Federal Republic of Germany* and for East Germany *German Democratic Republic*. I use the common names since they are still used to refer to the respective parts of Germany after reunification.

[6] Appendix A provides information on competing programs and how the unique characteristics of the policy change in 1999 can causally identify the effects of tax credits.

firms.[7] Tax credits typically reduced investment costs by around 10% but depending on the exact location and firm size, the reduction could be as high as 27.5%.[8] Tax credits were fully paid even if they exceeded tax liabilities of a firm and did not depend on the life span of the investment good.[9]

Importantly, all details on firm eligibility and the tax credit rate were precisely defined by law without room for discretion on a case by case basis. This led to an entitlement to tax credits for eligible firms and thus certainty for the planning of long-term investment projects.[10] The tax credits therefore can be considered a pure capital cost reduction for firms. The administrative cost for receiving tax credits was small. Firms filled out a tax credit claim form describing the investment good and the value of investment. Tax officers checked the correctness of the claim after the end of the business year and a positive assessment would trigger the transfer of tax credits. To reduce adverse incentives, a number of further eligibility criteria needed to be satisfied. Assets had to stay within the firm for at least 3 years to prevent East German firms from becoming pass-through companies of buying and reselling fixed assets.[11] Planes, passenger cars and low value assets such as office equipment or basic tools were never eligible because monitoring their continued operation within an eligible firm in East Germany (or West Berlin) would entail large costs.

Figure 2 summarizes overall government expenses for the tax credit program by year based on available information in the official subsidy reports. From 1992 to 1995 expenses totaled around €2 billion per year. After 1993, expenses declined steadily, which can be explained by the reduction in eligible industries. They reached a low of €645 million in 1998 and stabilized thereafter at around €1 billion per year. Starting in 2000, expenses from tax credits for investments in structures contributed to the total and generally made up around 15% each year.

In the empirical analysis, I focus on manufacturing firms as the main recipients of tax credits and consider the time period between 1995 and 2004, comparing their behavior before and after an important change in the program to tax credit rates for equipment investments in 1999. Starting in July 1994 manufacturing firms with at most 250 employees received a tax credit rate of 10% on equipment investments. Firms with more employees

[7] Retail businesses continued to have limited eligibility until 2001. Manufacturing service businesses such as construction design or research gained access to tax credits in 1999. Accommodation businesses were eligible from 2007.

[8] The highest rate applied to equipment investments of manufacturing firms with at most 250 employees in regions close to the Czech and Polish borders from 2002 to 2009.

[9] This is in contrast to special depreciation allowances where the decrease in cost of capital depends on the profit situation of a firm and on the years of depreciation due to the present value of future tax deductions. House and Shapiro (2008) and Zwick and Mahon (2017) provide detailed explanations.

[10] This is in contrast to various place-based policies that distribute grants to investment projects via a competitive application process with the final outcome based for example on the perceived success of the project or a commitment to hire additional employees.

[11] After 1999 the minimum time period was extended to 5 years.

instead received 5%. The program defined firm size as the number of employees at the beginning of a business year without differentiating full-time and part-time employment. Vocational trainees were excluded from the definition since their employment is considered educational and is governed by special contracts. The relevant tax credit rate applied to all investments of the same business year. In 1999, tax credits were raised for a broad range of equipment investments, with firms below the employment cutoff now receiving a rate of 20% and firms above the cutoff receiving 10%.[12] The announcement of the policy update was published in August 1997 (Bgbl, 1997 Part I/59). Because of disputes with EU law, the update got approval only by the end of 1998 (European Commission Press Release IP/98/1075). This adjustment led to higher tax credits for all manufacturing firms and thus a decrease in capital costs. However, firms below the cutoff were favored with an additional increase of 5 percentage points which translates into a larger relative decrease of capital costs of equipment for those below the cutoff.

The change applied to so-called modernization investments, which included among others any investment that could potentially increase production, change the production process or produce different products. Any investment in new equipment that did not directly replace a similar asset fell within this category. Even (high value) office equipment could be part of this category as long as it was bought in connection to a specific modernization investment.

Figure 3 summarizes the general tax credit changes for equipment investments of manufacturing firms in East Germany between 1995 and 2004.[13] Apart from the adjustment in 1999, there was another increase for modernization investments in 2000 that further strengthened the relative advantage of firms below the cutoff to those above. Tax credits for non-modernization equipment investment remained unchanged during the policy update in 1999 but were reduced in 2002. However, this reduction did not change the differential treatment and maintained a higher rate of 5 percentage points for firms below the cutoff before and after the change. [14]

Since the definition for firm size changed markedly in 2005, I exclude those years from my analysis. From then on the cutoff value followed the definition of small and medium firms by the European Union that takes ownership structure into account and defines the cutoff with respect to the number of employees, revenue and total assets.[15]

[12] At the same time an investment limit for receiving the higher tax credit rate of €2.56 million per year for firms below the employment cutoff was eliminated.

[13] Berlin had generally lower rates throughout the time period. Regions close to the Polish and Czech border received slightly higher rates between 2001 and 2009. For long investment projects, additional rules applied around changes of the tax credit rate.

[14] The tax credit rate for investments in structures was independent of firm size and thus, the reduction in capital costs of structures the same.

[15] The definition can be found in the "Commission recommendation of 6 May 2003 concerning the definition of micro, small and medium-sized enterprises" (ABl.EU, #L 124 pp36).

2 Theoretical Framework

To understand firm behavior after capital cost changes, it is helpful to outline a simple model of firm production. The literature has already developed detailed models of firm investment behavior in which adjustment costs and corporate taxation play an integral part.[16] Since I rely on a reduced-form approach in the empirical analysis, I abstract from both these issues and focus on a static model with capital costs consisting of a universal capital rental rate and a firm-specific tax credit rate.[17] Compared to these models, I focus on labor as input to the production process and consider heterogeneous labor types. To capture the influence of spillovers in regional firm production, I add a regional productivity shifter similar to Greenstone et al. (2010) and Gathmann et al. (2017).[18]

The model assumes many firms i within many regions r. Each firm produces one differentiated good according to the nested CES production function

$$F(K_i, U_i, S_i) = Y_i = A_i A_{ir} \left[(a_K K_i^{\rho} + a_S S_i^{\rho})^{\frac{\mu}{\rho}} + a_U U_i^{\mu} \right]^{\frac{1}{\mu}}, \tag{1}$$

where output Y_i is produced using capital K_i, low-skilled labor U_i and high-skilled labor S_i as inputs. The nesting of the three input factors follows Krusell et al. (2000) to allow for differential adjustment of the two labor inputs to a change in capital costs. In particular, the elasticity of substitution between low-skilled labor and capital (or high-skilled labor) is $\frac{1}{1-\mu}$ and the elasticity of substitution between high-skilled labor and capital is $\frac{1}{1-\rho}$. Production also depends on a firm-specific production parameter A_i and a productivity shifter A_{ir}. The productivity shifter is also firm-specific but additionally depends on aggregate outcomes in region r. I consider the behavior of A_{ir} in detail once I turn to the effect of spillovers.

Each firm chooses inputs according to the rental rate of capital r, wage w_U for low-skilled and wage w_S for high-skilled labor. I assume fully elastic labor supply and capital supply which leads to equalization of input prices throughout the economy. Firms take the input prices as given. The cost of capital still differs between firms since there is a firm-specific reduction through tax credits with rate τ_i. Firms set the product price p_i facing monopolistic competition with a downward sloping inverse demand curve

$$p_i = B Y_i^{-\frac{1}{\eta^D}}, \tag{2}$$

[16] Important examples are Hall and Jorgenson (1967), Hayashi (1982) and Abel and Eberly (1994). Hassett and Hubbard (2002) and Bond and Reenen (2007) summarize basic model assumptions and survey further approaches.

[17] The qualitative equilibrium results hold true nonetheless. An important difference is that in my model firms choose capital stock and not the investment rate as is the case with adjustment costs.

[18] Details on solving the model are provided in Appendix B.

where the price depends on the elasticity of demand $\eta^D > 1$ and a demand shifter B.[19]

The profit maximization problem for a firm is well defined and the first-order conditions fully explain the production decisions of a firm. The first-order conditions are

$$(1-\tau_i)r = \left(1 - \frac{1}{\eta^D}\right) B a_K Y_i^{1-\mu-\frac{1}{\eta^D}} X_i^{\frac{\mu-\rho}{\rho}} K_i^{\rho-1} (A_i A_{ir})^{\mu} \tag{3}$$

$$w_S = \left(1 - \frac{1}{\eta^D}\right) B a_S Y_i^{1-\mu-\frac{1}{\eta^D}} X_i^{\frac{\mu-\rho}{\rho}} S_i^{\rho-1} (A_i A_{ir})^{\mu} \tag{4}$$

$$w_U = \left(1 - \frac{1}{\eta^D}\right) B a_U Y_i^{1-\mu-\frac{1}{\eta^D}} U_i^{\mu-1} (A_i A_{ir})^{\mu} \tag{5}$$

where $X_i = a_K K_i^{\rho} + a_S S_i^{\rho}$.

To show the impact of a change in the cost of capital through adjustments in the tax credit rate on optimal capital and labor, I totally differentiate the production function (1) and all first order conditions (3), (4) and (5). By reformulating the results, the price elasticity of capital and the cross-price elasticity of high-skilled and low-skilled labor respectively are

$$e_{K_i} = \left[\eta^D s_{K_i} + \frac{1}{1-\mu} s_{U_i} + \frac{1}{1-\rho} s_{S_i} - \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right) s_{U_i} \frac{a_S S_i^{\rho}}{X_i}\right] Z \quad + \quad (\eta^D - 1) e_{A_{ir}} \tag{6}$$

$$e_{S_i} = \left[\left(\eta^D - \frac{1}{1-\rho}\right) s_{K_i} + \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right) s_{U_i} \frac{a_K K_i^{\rho}}{X_i}\right] Z \quad + \quad (\eta^D - 1) e_{A_{ir}} \tag{7}$$

$$e_{U_i} = \underbrace{\left(\eta^D - \frac{1}{1-\mu}\right) s_{K_i} Z}_{\text{direct effect}} \quad + \quad \underbrace{(\eta^D - 1) e_{A_{ir}}}_{\text{indirect effect}}, \tag{8}$$

where each elasticity $e_{\#} = \frac{d\#}{d\tau_i} \frac{1-\tau_i}{\#}$ is with respect to the tax credit rate,[20] $s_{K_i} = \frac{(1-\tau_i)r}{p_i} \frac{K_i}{Y_i}$ is the share of capital cost (after tax credit) in revenue, $s_{U_i} = \frac{w_U}{p_i} \frac{U_i}{Y_i}$ is the share of low-skilled labor cost in revenue, $s_{S_i} = \frac{w_S}{p_i} \frac{S_i}{Y_i}$ is the share of high-skilled labor cost in revenue and $Z = \frac{\eta^D}{\eta^D - 1}$ is an additional scaling term.

Each elasticity consists of a firm-specific direct effect and an indirect regional effect. Turning first to the direct effect of tax credits, the elasticities are similar to a model of firm production with just one labor type (e.g. Hamermesh, 1993). The elasticities of capital and labor largely depend on two effects, a scale effect from changes in input prices and a substitution effect between capital and the two types of labor. In the case of capital, these effects work in the same direction. An increase in tax credits leads to more demand for capital because of an expansion of production and a shift from labor towards

[19] Monopolistic competition leads to decreasing returns to scale. In this case, there is a unique interior solution. Other approaches introduce a fixed input factor or restrict full elasticity of labor supply.

[20] It is straightforward to show that this elasticity is equivalent to $-\frac{d\#}{dc_K} \frac{c_K}{\#}$, the capital cost elasticity.

capital. On the other hand, for each labor type the overall effect is ambiguous since there is higher demand from expansion in production but lower demand from the shift towards capital. The net effect depends on the elasticity of product demand and the elasticity of substitution of the relevant labor type. Because of monopolistic competition and the nesting of the CES production function, additional terms show up. First, less than fully elastic product demand amplifies adjustments of all inputs through the term Z. Second, for the elasticity of capital with respect to the tax credit rate, an additional term implies a change in the weighting of each substitution effect. Third, for the elasticity of high-skilled labor with respect to the tax credit rate, there is a dependence on the elasticity of substitution of low-skilled labor. Independent of the parameters, it is the case that the percentage change for capital is always larger than the percentage change for high-skilled or low-skilled labor.

To better judge the effect on the composition of labor inputs, it is useful to consider the elasticity of the ratio between high-skilled and low-skilled with respect to the tax credit rate. It is

$$\frac{d\frac{S_i}{U_i}}{d\tau_i}\frac{1-\tau_i}{\frac{S_i}{U_i}} = \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right)\frac{a_K K_i^{\rho}}{X_i}. \tag{9}$$

The sign and magnitude of this elasticity depend on the relative magnitude of each elasticity of substitution. If the elasticity of low-skilled labor is higher than the one of high-skilled, there is capital-skill complementarity with a shift towards high-skilled labor.

Turning to the indirect effect, a change in the regional productivity shifter influences each elasticity of input equally. Through the productivity shifter, the indirect effect even applies to firms without change in their capital costs. The elasticities of firm j with respect to the tax credit rate of firm i are

$$\frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} = \frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j} = \frac{dU_j}{d\tau_i}\frac{1-\tau_i}{U_j} = (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \tag{10}$$

The productivity shifter is supposed to capture spillovers between firms. I follow the literature and assume that the productivity shifter depends on the overall economic activity within a region. I define

$$A_{ir} = \sum_{j \in S_r} Y_j^{\lambda_{ij}}, \tag{11}$$

where the set S_r contains all firms in region r and λ_{ij} is the elasticity of agglomeration between firm i and j.[21] This definition encompasses many of the characteristics of spillovers discussed in the literature. Spillovers seem to be localized, governed by the proximity of firm production and labor inputs and aggregate output is a good measure to capture such

[21] I use output as the measure for economic activity since it is unambiguously greater or equal zero for tax credit rate increases.

economic density. Proposed mechanisms for the importance of proximity are cost savings in the supply chain of intermediate goods and services, knowledge spillovers and thick labor markets (e.g. Moretti, 2011). Apart from the importance of proximity these theories also suggest that not all firms profit equally. The firm-specific elasticity of agglomeration captures that benefits may vary because of the connections in the supply chain, similarity in production outputs and sharing of labor of different firms. An alternative explanation for spillovers are local multipliers (Moretti, 2010). An increase in local employment can lead to higher consumer demand of goods and services in the non-tradable sector in the same region and therefore higher employment in that sector.

To derive intuitive closed-form solutions, I simplify the productivity shifter by assuming an elasticity of agglomeration $\lambda_{ij} = \lambda$. The productivity shifter therefore becomes the same for all firms within a particular region.

With this simplification, the elasticity of regional productivity with respect to a change in tax credit rate of one firm is

$$\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \frac{\lambda}{\sum Y_j^\lambda}\sum Y_j^\lambda \frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j} = \frac{\lambda\eta^D}{1-\lambda\eta^D}\frac{s_{K_i}Y_i^\lambda}{\sum Y_j^\lambda}Z > 0, \tag{12}$$

where I assume that $\lambda\eta^D < 1$.

The elasticity importantly depends on the interaction of the elasticity of agglomeration and the elasticity of product demand. Larger values for both elasticities imply a more pronounced impact on regional productivity. Furthermore, firms with a higher share of output in total regional output and larger capital cost shares impact regional productivity more. An increase in the tax credit rate always leads to a positive spillover effect. The additional assumption on the parameters rules out boundary cases in which spillovers are so large that total output within a region becomes infinite.

If several firms receive a change in tax credits, then the effect on regional productivity gets larger since more firms increase their output through the direct channel. Intuitively, the elasticity with respect to the tax credit rate of multiple firms comes about by sequentially calculating the equilibrium adjustments. Considering infinitesimally small tax rate changes, this reduces to summing up all elasticities of regional productivity for firms with changing capital costs. For further notational simplicity, I consider only the case where firms start out with the same tax credit rate $\tau_i = \tau$ and a subset receives the exact same tax credit rate change $d\tau_i = d\tau$. As result, the elasticity is

$$\frac{dA_r}{d\tau}\frac{\tau}{A_r} = \sum_{i|d\tau_i\neq 0}\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \frac{\lambda\eta^D}{1-\lambda\eta^D}Z\frac{\sum_{i|d\tau_i\neq 0} s_{K_i}Y_i^\lambda}{\sum Y_j^\lambda}. \tag{13}$$

The elasticity has a very similar form to the one before. However, the adjustments of multiple firms lead to a summation of the output of all firms with tax credit rate change

weighted by their share of capital costs. This means that firms get a larger indirect effects from spillovers in regions where more firms receive a tax credit rate change. This effect increases output and all input factors independent of the tax credit rate of a firm.

In sum, the theoretical framework predicts the following. An increase in the tax credit rate leads to the use of more capital and to higher output. The sign of the effect of labor depends importantly on the magnitude of the substitution effect between capital and the two labor types. Whether there is a shift towards high-skilled or low-skilled labor depends on the relative magnitude of substitution between each labor type and capital. On top of these direct effects, spillovers between firms lead to an additional positive effect on capital, employment and output. This effect is larger in labor markets with more output in firms that experience an increase in the tax credit rate. Finally, the proposed theories for spillovers suggest that their influence within a region can vary firm by firm.

3 Estimation Strategy

The estimation strategy is guided by the theoretical framework and uses the described update of tax credit rates in 1999. I implement a difference-in-differences estimation approach and compare the adjustment of firms below and above the firm size cutoff before and after the change. I start by estimating average effects using the regression model

$$Y_{ibt} = \beta \text{Treated}_b \times \text{Post}_t + X_{it}^{'}\gamma + \psi_i + \psi_{nt} + \psi_{lt} + \epsilon_{ibt}, \tag{14}$$

where Y_{ibt} is the outcome variable for each firm i with treatment b in year t. The variable Treated_b classifies firms into treatment and control group. I consider firms with at most 250 employees in 1998 as treated and those above as untreated to reflect the relative advantage for firms below the cutoff. The classification is fixed over time. I interact this variable with the dummy variable Post_t, which categorizes years 1999 to 2004 as treatment period to reflect the change in tax credit rate. ψ_i, ψ_{nt} and ψ_{lt} are firm, industry-year and labor market-year fixed effects, respectively. Firm fixed effects control for level differences in firm characteristics that stay constant over time such as those correlated with average firm size, industry and location. Industry-year and labor market-year fixed effects can control for the possibility that industry-specific shocks, and labor market-specific policies and economic developments coincide with the update of the tax credit program. I include further control variables $X_{it}^{'}$ depending on the specification with log average firm wage being used for all main results.[22]

The coefficient of interest is β. Without confounding factors, it provides a causal estimate for the effect of the reduction of capital costs caused by the policy change on

[22] When using pre-treatment wage growth trends instead of log wage, results stay qualitatively unchanged.

each outcome variable. The set of fixed effects already controls for various confounding factors. However, it cannot take care of distinct firm incentives that are introduced by the firm size cutoff. Firms can adjust their size over time and thus are able to cross the cutoff. Such movements imply that these firms receive a different tax credit rate than assigned by the treatment status in the regressions. If this change in firm size is unrelated to the tax credit program, for example because of a general decline of a specific firm, this would not influence the causality claim, although it would influence the interpretation of the coefficients since it partly captures intent to treat effects. To minimize this issue, I exclude firms that are close to the cutoff in 1998 and thus those that are more likely to cross the cutoff in either direction. On the other hand, if firms intentionally move just below the cutoff or delay moving above, this would result in bunching at the cutoff. Apart from estimations with log employment as outcome variable, I further exclude observations in any year for which firm size is close to the cutoff to avoid including firms that bunch around the cutoff.

I study several outcomes. First, I am interested in the effect on capital inputs. As is common in the literature, I use investment as a directly measured variable in the dataset to proxy capital adjustment. I consider the intensive margin, extensive margin and a combined measure. I also specifically consider equipment investment. In a second step, I focus on labor inputs and analyze the effect on the number of total employees and for various subcategories separately. To assess the impact of the policy on the skill composition of a firm's workforce, I also use employment ratios by education and occupation as an outcome variable.

The error term ϵ_{ibt} includes all other omitted factors. I cluster standard errors on the regional level allowing for heteroscedasticity and arbitrary correlation between firms within the same region over time. I consider German *Landkreise* as regions in my analysis. There are 76 regions in East Germany with on average 13 firms in the empirical analysis.[23] The regions are then divided into 56 labor markets following a classification used by Dustmann and Glitz (2015).

Since the average effect can mask interesting adjustment patterns, I estimate the dynamic regression model

$$Y_{it} = \sum_{p=1995}^{1997} \delta_p D_{bp} + \sum_{p=1999}^{2004} \delta_p D_{bp} + X_{it}^{'}\gamma + \psi_i + \psi_{nt} + \psi_{lt} + \epsilon_{ibt} \tag{15}$$

where in comparison to specification (14) the interaction term is omitted and instead a set of yearly dummies D_{bp} is introduced. Each dummy variable assigns a value of one to firms in the treatment group for the corresponding year p and zero otherwise. The

[23] Various regions merged due to reforms on a state level. I use the regional disaggregation as of 2014 to ensure consistency over time.

coefficients δ_p for the years 1999 to 2004 capture the dynamic treatment effect. If the treatment effect is indeed causal, then treatment and control group have parallel trends absent the policy change. This would not be the case if anticipation effects from the announcement of the policy or long-term influences from similar policy changes before 1995 influence firm behavior. For this reason, I also examine the pre-treatment effect by including dummies for the years 1995 to 1997. Observing statistically insignificant estimates close to zero before the start of the treatment provides an indication that the identifying parallel trends assumption indeed holds true.

Both regression approaches so far only consider the direct effect of the policy change. For the estimation of spillover effects, I use the regression model

$$Y_{iblt} = \beta \text{Treated}_b \times \text{Post}_t + \alpha \text{ShareBelow250}_{l,98} \times \text{Post}_t + X_{it}^{'}\gamma + \psi_i + \psi_t + \epsilon_{iblt}, \quad (16)$$

where $\text{ShareBelow250}_{l,98}$ is the share of employees working in firms with at most 250 employees in labor market l for year 1998. The variable is interacted with the treatment period dummy Post_t to set up a difference-in-differences estimation with continuous treatment status. There are now two coefficients of interest, β, the direct effect of receiving a higher tax credit rate as in specification (14) and α. The latter coefficient provides an estimate of the difference in the outcome variable by the share of firms below the cutoff that is due to the change in tax credit rate. This setup mimics the results of the theoretical framework with α corresponding to an estimate of the spillover effects. Since I use variation on a labor market level for this estimation, I cannot control for the same set of fixed effects. Instead, I include firm and year fixed effects and the log of firm wage as baseline controls.

The analysis of dynamic effects is again helpful to better understand adjustment behavior and to check the parallel trends assumption. The regression model is

$$\begin{aligned} Y_{iblt} = \sum_{p=1995}^{1997} \delta_p D_{bp} + \sum_{p=1999}^{2004} \delta_p D_{bp} + \sum_{p=1995}^{1997} \theta_p \text{ShareBelow250}_{lp,98} \\ + \sum_{p=1999}^{2004} \theta_p \text{ShareBelow250}_{lp,98} + X_{it}^{'}\gamma + \psi_i + \psi_t + \epsilon_{iblt}, \quad (17) \end{aligned}$$

where $\text{ShareBelow250}_{lp,98}$ is the weighted share of firms with at most 250 employees in labor market l and year p. It is zero otherwise. The set of coefficients δ_p still estimates the direct impact of the policy change over time. The set of coefficients θ_p estimates the dynamic effect of spillovers over time. If the effect of the policy on spillovers is causal, then there should not be any differential effect between labor markets before the policy change. I check for this assumption by including coefficients for pre-treatment periods.

4 Data

4.1 Data Sources

The empirical analysis relies on two data sets, the AFID Establishment-Panel by the Federal Statistical Office of Germany and the Establishment History Panel (BHP) by the Institute for Employment Research (IAB).

The AFID dataset has a broad coverage of variables for investment, employment and output for the universe of manufacturing and mining firms with more than 20 employees in Germany. With its unusual richness it perfectly fits the needs for the general empirical analysis. Firm variables are collected through various administrative surveys and are used to inform the government and the public about key economic statistics like aggregate output and investment. Because of the importance of these statistics, firms are required by law to provide truthful information. The AFID dataset merges the underlying surveys through a unique firm identifier and aggregates information from monthly and quarterly surveys by year. Information is available since 1995 and new waves are continuously added. The dataset is especially suited for the investment analysis since there is separate information for equipment. Further subcategories distinguish different modes of acquisition such as self-production, leasing and purchase. For measures of output, there are revenue, production value, orders and the number of distinct products. Revenue is divided into domestic and foreign. There is however only limited information on labor inputs, with total employment and wage bill being most informative.[24] For each firm the 4-digit industry code and location on a regional level is provided as well.

For a more detailed employment analysis, I use the BHP dataset which provides information on overall employment, employee composition, employee inflows and outflows, and wages. The data are based on the employment histories of the entire labor force covered by social security. They are collected from mandatory communication between firms and the Federal Employment Agency on changes in employment. The BHP aggregates this information on a firm level for 30 June of every year for West Germany since 1975 and for East Germany since 1992. I focus on the years 1995 and 2004. The final sample consists of a 50% random draw of firms and all available years of selected firms are included. For employment there are counts for the total and for subcategories of education, occupation types and age and full-time employment and vocational training. For a better understanding of changes over time, inflow and outflow information provides the number of employees that did not work in the same firm one year before and one year after, respectively. These flows are again divided into subcategories. I use information for incoming and leaving employees that were unemployed one year before

[24] There is a distinction of employees by contract type. I do not use this information since it does not translate well to other economic concepts.

and after, respectively. Average wage is based on full-time employees and available for quartiles and by education. The dataset includes firms of all industries. Information on a 3-digit industry level and a region variable allows the selection of relevant firms.

The AFID and BHP dataset are distinct and it is not possible to link them. Therefore, it is necessary to calculate key policy variables separately and take into account changes in the data collection over time independently. I explain the general adjustments here but relegate further details to Appendix C.

One important variable in the analysis is firm size, since it is the basis for classifying treatment and control group. The program definition requires information on the head count of overall firm employment and vocational trainees.[25] The AFID dataset lacks information on the latter. To address this issue, I match the number of vocational trainees on a firm-level for the years 1999 to 2001 from the cost structural panel (KSE) by the Federal Statistical Office.[26] For observations that are unmatched, I impute values assuming a constant share of trainees within firms or if unknown within industries. For the BHP dataset, on the other hand, information on the marginally employed is missing for years before 1999. I impute missing observations by assuming a constant share of marginally employed within firms or if unknown predict the share within industries for different firm size.[27] The change in reporting of marginally employed unfortunately coincides with the policy change. To reduce the risk that the imputation of marginally employed influences the estimation, I drop firms with an average share of more than 25%, which is above the 95th percentile within the manufacturing industry.

Another issue concerns the continuity of firm identifiers. In the AFID data, firm identifiers are constant even if the ownership structure of a firm changes, in the BHP this is not the case. However, in the BHP the firm identifier can change for relatively simple reasons such as changes in the legal structure. In the analysis, firm structure possibly influences decision making processes and thus leading to abrupt changes in production behavior. To have firm identifiers that exclude such changes, I separate firm identifiers in the AFID dataset when there are changes in overall firm structure and I exclude firms with more than one-hundred establishments in a given year. For the BHP, I follow Hethey-Maier and Schmieder (2013) and create unique identifiers for firms that are connected through employee flows.

Further adjustments are the reclassification of regions as of 2014, the classification of

[25]The definition considers the business year start for calculations. Employment figures in the AFID and BHP dataset do not have this same timing, however, an auxiliary analysis using the IAB establishment survey does not show a systematic difference of employment levels in manufacturing firms within a given year.

[26]The KSE is a yearly firm survey of a stratified random sample in the manufacturing and mining industry and focuses on the production process.

[27]Since I rely on broad firm size intervals in the analysis, and vocational trainees and marginally employed constitute a small share of a firm, measurement error from both imputations should lead to relatively few misclassifications into treatment and control group.

regions into labor markets according to Dustmann and Glitz (2015) and a classification of 2-digit industries that further aggregates uncommon industries.

4.2 Sample Selection

The sample selection follows the eligibility criteria of the program. First, I select firms in manufacturing industries[28] for years 1995 to 2004 located in East Germany excluding Berlin. To avoid peculiar behavior of entering and exiting firms, I also condition on them being economically active throughout the period of the analysis.

Second, I check for bunching behavior around the cutoff. Figure 4 displays the size distribution of manufacturing firms in East Germany and West Germany around the cutoff for 1999 to 2004. For West Germany a decrease in the density with increasing firm size is apparent which is as expected (e.g. Axtell, 2001). For East Germany this pattern is generally true as well, however, just below the cutoff there is excess mass. This points to bunching of East German manufacturing firms. I therefore exclude all firms with a size of between 226 and 274 employees in 1998.[29] I further exclude observations within that size interval in any other year.[30]

As a last step, I include only firms with at least 40 employees and a maximum of 1,500 in 1998 and observations that lie within the same interval for any other year. This reduces the problem from biases due to heterogeneous effects among observationally different firms (Heckman et al., 1997).[31] To check whether the choice of the excluded interval and the size interval has an impact on estimation results, I run robustness test that vary these interval boundaries.

Figure 5 considers the impact of the exclusion around the cutoff on firms moving outside their assigned treatment status. As a comparison, I include the case without restriction. The specific sample selection has little impact on the treatment group. At most 3.5% of firms move above the cutoff and once observations around the cutoff are excluded this share reduces to a maximum of 1.5%. There is more movement within the control group. Without the exclusion of firms around the cutoff, 8.6% of firms in the control group fall below the cutoff already in 1999. In 2003 this share is at 26.4%. It is likely that some of these firms move on purpose to take advantage of higher tax credit rates. For the sample with excluded bandwidth, the share of firms moving below is considerably lower. After the policy change the share increases slowly over time,

[28] These include all industries with a WZ 1993 classification of D.

[29] I determine this cutoff by implementing a structural approach that adapts ideas from Garicano et al. (2016). The maximum likelihood estimation leads to a value of 274.6. Implementation details are available upon request.

[30] The exclusion of single observations is not appropriate for estimating the effect on employment since then there would be a selection based on the dependent variable. In this case, I only condition on firm size in 1998.

[31] I implemented a propensity score matching approach between treatment and control group and found qualitatively similar results to those in the main text.

reaching 6.8% in 2001 and is highest in 2004 with 16.9%. Even though the share in 2004 is non-negligible, in this case, a movement below the cutoff is less problematic since these observations are not affected by bunching behavior.

4.3 Descriptive Statistics

Table 1 presents a selection of firm variables from the AFID dataset in Panel A and the BHP dataset in Panel B for the years 1995 to 2004.[32] I show descriptive statistics for all manufacturing firms in West and East Germany (excluding Berlin), and for treatment and control group of the empirical analysis. Overall, firms in West Germany are larger in many respects compared to those in East Germany. The average number of employees in the AFID dataset is 155.19 in West Germany and 89.42 in East Germany. They have the same likelihood of investing in any given year but in West Germany the investment value is larger. These differences in input factors translate to higher revenue with €30.75 million compared to €13.03 million. Panel B reports very similar differences for the number of employees. On top it shows that full-time employees earn considerably more in West Germany. In terms of employee composition, there are actually more employees with college degree, more high-skilled occupations and more vocational trainees in East German manufacturing firms.

Turning to the estimation sample, there are again clear differences in size. This is not surprising given the definition of treatment and control group. The treatment group is similar to the average East German firm when comparing means, but the control group is far larger in every respect.[33] The groups have similar employee composition. For example the share of college graduates is 11.16% in the treatment group compared to 14.82% in the control group.

As a step towards the actual estimation, Figure 6 presents the raw means for treatment and control group over time for different investment and employment outcomes. Encouragingly, in each plot treatment and control group have a similar pre-treatment trend. In the upper left panel, there is a continuous decrease in the log of total investment that continues after the policy change in 1999. For subsequent years the investment level stays higher for the treatment group reflecting a positive investment response. The decision to invest, shown in the upper right panel, is relatively stable with a value close to 100% and there is little differential movement before or after 1999. I plot log employment using AFID data in the lower left panel and BHP data in the lower right panel. The evolution in both graphs is quite similar. This is remarkable given that the actual data collection was independent of each other and speaks to the quality of both datasets.

[32] I provide descriptive statistics for the complete list of relevant variables for the estimation sample in the Appendix.

[33] This does not mean that there is no overlap. Standard deviations are usually large, especially for investment and employee composition.

Until 1999, employment increases for the treatment and control group. Subsequently, employment stays constant or decreases in the control group whereas there is continued growth for the treatment group until 2001. After 2001, employment decreases for the treatment group as well at similar levels as the control group. This pattern indicates a positive employment response to the relative reduction in capital costs.

5 Results

5.1 Investment

As a first outcome I study firm investment. Table 2 summarizes the average effect for several investment measures. In column (1), I consider log of total investment at the intensive margin and find that the policy change leads to 23.4 log points higher investment among firms below the firm size cutoff. Since the policy change only affected tax credits on equipment investment, column (2) reports the estimate for this subcategory. The estimate is slightly higher with 25.1 log points and statistically significant. Both outcomes measure only the intensive margin. In column (3), I check for differences in the probability of investing. However, there does not seem to be any response with a point estimate of zero and small standard errors. This is likely the case because of the high rate of firms that invest in any given year independent of treatment. In the literature, one outcome of interest is the investment rate (I_t/K_{t-1}) which combines intensive and extensive margin. I do not observe capital in my datasets. As proxy, I consider investment divided by the average total investment during the period of analysis. I find a positive and statistically significant response of 0.171 for total investment and 0.167 for equipment investment.

Taken together, these results show a positive average investment response from tax credits. In order to compare the magnitude to previous findings in the literature, it is possible to calculate the elasticity with respect to capital costs. I calculate the terms $\Delta t_i/(1-t_i)$ for treatment and control group taking into account the further tax credit rate increase in 2000 and assuming that all investments are modernization investments. Capital costs decreased by 15.74% for the treatment group and 7.46% for the control group. This leads to an intensive margin semi-elasticity of total investment with respect to capital costs of 2.825.[34] This elasticity can be considered large. The consensus range of elasticities proposed by Hassett and Hubbard (2002) is 0.5 to 1.0. Recent studies including House and Shapiro (2008), Maffini et al. (2016) and Zwick and Mahon (2017) find much larger elasticities of around 8. Zwick and Mahon (2017) provide evidence that smaller firms have a larger elasticity. This could explain my findings, since firm size is relatively small.

[34]For cases where there are structures and replacement investments, the calculated elasticity is a lower bound. The extensive margin elasticity of total investment with respect to the tax credit rate is zero.

Table 3 considers a few robustness tests for the investment specification. Column (1) reproduces the results from before. Given that investment is highly volatile, in column (2), I consider a sample where I exclude firms with investment growth in 1997 above the 95th percentile. The response is slightly lower for log of total investment and log of equipment investment but there is no response for the probability of investing. In column (3), I only control for average firm wage, firm fixed effects and year fixed effects. The coefficients are again smaller but lead to qualitatively similar results. Finally, in column (4), I only consider single-establishment firms to eliminate inconsistencies between tax credit eligibility and the level at which production decisions are taken. I again exclude firms with high volatility in their investments. In this case, the average effect becomes slightly larger. There is still no response in the extensive margin. Overall, the investment response is similar throughout quite distinct specifications and sample selection.

For a better understanding of investment behavior over time, I study the dynamic specification. Figure 7 summarizes the coefficients including 95% confidence intervals for the intensive and extensive margin. For both measures of the intensive margin, there is an upturn directly in the year after the policy change. In the subsequent year, investment further increases with coefficients of 20.6 log points for total investment and 23.2 log point for equipment investment. Afterwards, coefficients stabilize around 20 log points. There seems to be another increase in 2004. Standard errors are relatively large so that not all coefficients after the policy change are statistically significant. For pre-treatment periods, coefficients are close to zero which suggests that treatment and control group follow the same trend. For the extensive margin, there is not any clear dynamic pattern. Coefficients move around zero for periods before and after the policy change leading to the zero result on average.

5.2 Total Employment

Given the positive response in investment, it is then of interest whether there are changes in the use of labor inputs as well. Table 4 presents the estimation results for the effect of the tax credit rate change on employment within firms. For the regression of column (1), I use employment information from the AFID dataset. I find a positive and statistically significant effect of 11.3 log points on total employment. The response is smaller than for investment but still sizable. In column (2), I consider the same employment measure based on the BHP data. In this case the coefficient is 8.7 log points which is similar in magnitude and confirms the positive employment response. To check whether the employment effect applies to different subgroups of employees, in column (3) to (5), I consider regular employees, full-time employees and full-time regular employees.[35] The

[35] These measures largely exclude marginally employed and thus allow to check the influence of the change in reporting of marginally employed in 1999 by comparing it to total employment.

average effect is 7.6 log points for regular employees, 8.8 log points for full-time employees and 8.7 log points for full-time regular employees which is very close to the effect on total employment.

Figure 8 shows the according dynamic results for total employment and full-time employment. For the case of AFID data, there is a continuous increase in total employment during the treatment period. In the first year after the policy change employment is 3.3 log points higher. After three years it reaches 9.9 log points and then levels off. For the case of BHP data, the effect on total employment is more immediate. After one year, employment is 7.7 log points higher. Subsequently there is a small drop, that is followed by a slow increase over time reaching 10.9 log points in 2004. When considering only full-time employees, there is again an immediate response of 5.5 log points. Even though there is again a drop in the subsequent year, there is a stronger increase in the effect over time, reaching 12.3 log points in 2004. When checking for parallel trends in employment for pre-treatment periods, coefficients are again close to zero and mostly statistically insignificant. For total employment with the AFID data there seems to be some movement already in 1998, however, it is small in magnitude.

One concern for the employment estimations is the slightly different sample selection of observations around the cutoff, and at the lower and upper bound. Since for employment I only condition on firm size in 1998, there is the possibility that firms bunch around the cutoff in later years and bias the estimation. For this reason, I consider the yearly change in the log of total employment over time in Figure 9 using BHP data. In this case, I follow the standard sample selection excluding firms by size in 1998 and observations according to size in any other year, which eliminates bunching observations. In years before the policy change, the effect is close to zero and statistically insignificant. In 1999, there is positive growth in total employment of 7.9 log points. In subsequent years coefficients are statistically insignificant again. They are still in line with the level results suggesting a decrease in employment in 2000 and small positive growth over time in subsequent years.

5.3 Flow Information

I continue with an analysis of employee flow information. The BHP data reports the number of employees within each firm that did not work there one year prior or that do not work there one year after. For my empirical analysis, I first relate the inflow and outflow information to overall employment one year prior. This is equal to the yearly growth rate for net flows and the hypothetical growth rate for inflows and outflows assuming the other flow value to be zero. I then cumulate these rates over time and take logs to have a measure which is similar to log employment.[36] I apply the same procedure to flow information of employees within firms that were unemployed one year prior or are

[36]This is the case since firm fixed effects control for average firm employment.

unemployed one year after.

Table 5 reports the results for each of the cumulated flow variables. In column (1) to (3), the dependent variables include flows from all employees. The result for net flows in column (1) is nearly identical to the result using log employment with an increase of cumulated net flows by 8.6 log points. This serves as a check on the validity of the analysis of flow information. In column (2), I focus on the inflow rate and find an increase by 10.2 log points. Since the additional inflows in treated firms are larger than net flows, there must be an increase in outflows as well. This is confirmed in column (3), although the coefficient is small and statistically insignificant. These results show that firms that received relatively more tax credits after the policy change increased firm size predominantly through hiring additional employees. Importantly, the point estimate does not suggest that the control group has higher outflows which counters the concern that the employment effect is due to a shift of employees from the control group to the treatment group. If anything, the higher cumulated outflow among treated firms suggests changes in employment structure.

From a policy perspective, it is of interest whether unemployed individuals gain from tax policies. I analyze the flows related to unemployment in column (4) to (6). Even though the coefficients are statistically insignificant, their magnitudes are economically significant. The effect of inflows from unemployment in column (5) is 5.0 log points. When comparing this estimate to the one for overall inflows, the hiring of unemployed people constitutes 49% among the additionally hired employees. This is sizable and similar to the share of all hires of 60%. The effect on the cumulated outflow rate into unemployment is 1.8 log points which is almost as high as for overall outflows. Thus, it seems that the additional separations are of employees who have problems finding re-employment.

I consider the dynamic effects of the flow variables in Figure 10. Taken together, the dynamic results confirm the previous findings. It is of interest that the net flow and inflow for unemployment are statistically significant in this specification. It is also the case that the share of inflows from unemployment is larger at the beginning with 71.4% and reduces over time. Furthermore, the increase in outflows, although still statistically insignificant, slowly increases over time which suggest that firms first hire additional employees and only let go of unnecessary employees over time.

5.4 Capital-Skill Complementarity

In the literature, there is an important discussion about capital-skill complementarity. I examine this adjustment mechanism in the context of tax credits by analyzing firm ratios of high-skilled labor, high-skilled occupations and manual occupations. I classify the skill level of labor according to education and distinguish between college educated and others. I build the same ratio based on full-time employees to have a measure independent of pos-

sible influence by the marginally employed. For the classification of different occupations, I build on the categories provided in the BHP dataset. I consider technicians, engineers, semi professions, professions and managers as high-skilled occupations. As another dimension, I am interested in the shift of manual labor. These are occupations that are either low-skilled or medium-skilled. I compare them to low-skilled and medium-skilled service and sales occupations. Finally, within manual labor, I compare medium-skilled (qualified) to low-skilled occupations. Table 6 reports the average effect on each ratio. They are all close to zero and statistically insignificant. Given that the estimate is in log points of a ratio, I can reject large changes in employment composition. For example, for the ratio of college educated the upper bound of the 95% confidence interval is 5.2 log points which would shift the high-skilled labor ratio just from 0.131 to 0.138.

This zero result is surprising given the large literature concerning aggregate changes in inequality and job polarization. The literature considers ICT as an important mechanism for these shifts, typically found starting in the 1990s. It is possible that in my setting, ICT does not play a big enough role and therefore does not influence the average employment composition. This however does not mean that there is no influence in particular circumstances. To explore this issue in more detail, I study heterogeneous effects of tax credits by the intensity of ICT usage. Since the main datasets do not have information on this type of capital, I match various data on an industry level. For the analysis of composition changes by skill, I rely on information of industry-specific ICT usage from several data sources. Sauer and Strobel (2015) provide investment information for 2014 based on data by the Federal Statistical Office and a firm survey by the ifo institute. The EU KLEMS project, in collaboration with the DIW, provides various capital input measures since 1991 for Germany. I rely on the real fixed capital stock in 1998. The IAB establishment survey includes extensive margin investment decisions on firm level for 1993-2014. I aggregate this information to 2-digit industries after controlling for broad regional areas, firm size, average wage and year as possible confounding factors. Finally, the Economic Census collects capital expenditure information for U.S. manufacturing firms. I select year 2002 as it is the earliest publicly available one.

Table 7 presents the estimation results using various measures on ICT. I find that firms in industries with higher ICT shares change their employment composition more towards high-skilled labor and high-skilled occupations when receiving investment tax credits. This heterogeneous response is remarkably stable for each ICT measure, even though they are from quite distinct sources. The effect on manual occupations and on shifts within manual occupations is noisy. Depending on the measure for ICT, the response is either negative or positive. Nevertheless, using information from the Economic Census, I find a statistically significant effect for a larger shift towards qualified manual labor among industries with higher ICT shares.

5.5 Revenue

Because of a higher tax credit rate, firms increase investments and employment. It seems natural to investigate whether at least some of this effect turns into higher output. Table 8 provides estimation results for various revenue measures and the number of distinct products as proxy for output.[37] In column (1), I use total revenue as outcome variable and find a small effect of 1.1 log points. However, for domestic revenue in column (2), the effect is 8.3 log points and for domestic revenue of only manufacturing-specific goods in column (3), it is 8.0 log points. This is puzzling since there must be a reduction in exports to compensate for changes in domestic production, even though exports make up a small share of production on average. I therefore consider the possibility that the results for total revenue are driven by outliers among exporting firms. In column (4), I restrict the sample to observations with a maximum average export share of 15%. In this case, the effect is 7.6 log points which is close to the effect on domestic revenue (though statistically insignificant). Thus, the discrepancy between the effect on total revenue and domestic revenue seems to be confined to those firms that already export sizable amounts. As additional outcome I include the number of distinct products in column (5) but I do not find an effect of tax credits on the number of products.[38]

5.6 Robustness Tests

Taken together, these first results provide a positive assessment for investment tax credits. Not only do they increase firm investment, but they also lead to more employment overall and for various subgroups, and thereby to higher revenue. The results are all based on a specific selection of the firm size sample. I check how stable these results are when varying on the one hand the interval of excluded firms around the cutoff and on the other hand the interval of smallest and largest included firms. Furthermore, the estimation relies on the difference in tax credit rate between firms below and above the cutoff as the defining margin. I test this behavior by using several policy irrelevant cutoffs within the treatment group which should lead to zero results.

Table 9 reports coefficients for the most relevant measures of investment, labor and revenue. The interval of excluded firms around the cutoff varies in each column. In column (1), I do not exclude any firm. Column (3) follows the main specification with an excluded interval of 225 to 275 employees. In addition, I report an intermediate case in column (2) and larger intervals in column (4) and (5). Reassuringly, estimates for all outcomes are robust throughout the different intervals. This applies especially to log overall investment, the extensive margin investment decision and employment. For log equipment investment there are slight differences although there is no clear pattern as

[37] The revenue measures mask the actual output response by possible changes in output prices.

[38] I cannot check whether firms introduce novel products and at the same time discontinue old ones.

a function of the size of the excluded interval. For log domestic production, coefficients grow somewhat by the size of the excluded interval, but qualitatively, the results are the same. Taken together, there is little evidence that the exclusion of firms around the cutoff has a notable impact, possibly because the number of firms that bunch at the cutoff is small in comparison to the whole sample.

Table 10 keeps the excluded bandwidth constant but instead varies the smallest and largest included firm size. In this table, column (4) reproduces the main specification and columns to the left use smaller intervals whereas columns to the right use larger intervals. Again, independent of the chosen interval, estimates lead to qualitatively similar results with limited fluctuations in the main coefficient of interest. The largest differences apply to the investment variables for which the coefficients in column (1) are somewhat smaller and those in column (6) larger than the rest. It should be noted that for the smallest firm interval, the number of observations is considerably smaller which may lead to higher susceptibility to outliers.

Finally, Table 11 reports the estimation result for the placebo cutoffs. The sample only consists of firms in the treatment group. I focus on the cutoffs 80, 100 and 125 to have a reasonable number of firms below and above these new cutoffs. I still exclude firms around the cutoffs as in the main specification. For the majority of outcomes, coefficients are close to zero and statistically insignificant. This holds true for all placebo cutoffs. For the extensive margin investment decision, coefficients are negative and statistically significant. However, given the high probability of firms investing in any given year, the coefficient is not economically significant.

5.7 Spillover Effects

All results so far speak to the direct effect of investment tax credits. I now exploit the regional variation in the number of firms below the cutoff and thus variation in the number of firms that benefited more from the policy change to examine spillover effects.

I start with an analysis that does not consider differences in treatment intensity through linkages between industries. Table 12 reports the results according to specification 16. I also show modifications that vary the type of fixed effects included in the regression and the firm size interval. The table includes estimates for the direct effect and the spillover effect. I find a consistent direct effect of tax credits on employment. For the main specification in panel 1 column (1), the coefficient is 0.081 which is similar to the previous regressions. For all other specifications the direct effect is quite similar. The indirect effect is not as robust. In the main specification, the coefficient is 0.272, statistically significant and stable when including industry-year fixed effects and area-year fixed effects. However, when increasing the firm size interval, the magnitude of the coefficient drops considerably. With a lower firm size boundary of 10 employees, the coefficient is

only 0.054.

To better understand the differences in these coefficients, I plot the dynamic spillover effect for firms with at least 20 and at most 1,500 employees in 1998 in Figure 11. The spillover effect appears immediately after the change in the policy. However, there is a further sizable increase in the effect over time, which suggests that the propagation to other firms does not happen immediately. For pre-treatment periods, the spillover effect is close to zero and statistically insignificant, suggesting parallel trends between firms in different labor markets.

It is a common view, that the magnitude of spillovers varies within a region depending on how benefits from one firm propagate to others. One propagation mechanism considers input and output linkages between industries. I examine this mechanism by splitting the regional share of firms below the cutoff into firms that are within the same industry and those in another industry. If input-output linkages play a role then the share of firms in the same industry should have a stronger effect. Table 13 reports the results. In the majority of specifications, it is indeed true that firms in the same industry have a stronger effect on the employment effect of a firm. It should be noted that many coefficients are not statistically significant. The results are therefore only indicative.

The policy was focused on manufacturing firms. It is possible that spillovers translate to benefits for the service industry as well. It is however less likely that such an effect happens through input-output linkages or knowledge spillovers. Nevertheless, it is possible that higher product demand on the consumer side leads to benefits through multiplier effects. Table 14 present the results for estimating specification 16 excluding the direct effect term from the regression. Independent of the firm size selection and chosen control variables, estimates are small and statistically insignificant. Therefore, spillover effects seem to be confined to firms in the manufacturing industry. Taken together, the results suggest that the similarity of firms is important in determining spillover effects and that multiplier effects do not occur.

6 Discussion

The results show that investment tax credits work in the intended way, however, it is important to relate the benefits to the incurred government expenses to let policy makers decide on the most efficient policy given their intended goals. For understanding the effectiveness of investment tax credits, one important measure relates the government expenses needed for increasing employment among the average manufacturing firm in East Germany. To better understand the contribution of spillovers, the measure can be calculated assuming only direct effects and including spillover effects.

I first provide a back-of-the-envelope calculation of government expenses for the aver-

age East German manufacturing firm based on the direct effect. Using information from Table 1, the average firm invests €1.17 million per year and over the sample period, there are on average 6,415 firms per year. As explained in the results section, the measured effects come from a change in the relative tax credit rate of 7.083 percentage points. At the intensive margin of investment this leads to an increase by 23.4 log points but no change at the extensive margin. On average 86% invest in a given year. Taken together and assuming a 10% tax credit rate, tax credits induce firms to invest on average 33 log points more, translating to €329.42 thousand more in investment. At the same time the average firm increases employment by 12.2 log points translating to 10.33 employees more. These improvements are achieved by providing €117.2 thousand in tax credits. Thus, government expenses of €11,346 lead to one additional employee.

When including spillovers, the calculations are as follows. If we assume a tax credit rate of 10% for all firms, then the share of employees working in firms that receive a tax credit is one. Based on column (3) in Table 12, this translates to a direct adjustment effect in employment of 15.54 log points and a spillover effect on employment of 19.5 log points. Thus, spillovers explain 55.6% of the overall effect and decrease government expenses for one additional employee by the same amount.

The calculated share of spillovers should be considered an upper bound. It assumes that all relevant firms receive tax credits. Many times the tax credits are only for a subset of firms as is the case for Germany where it was focused on the manufacturing industry. However, firms might rely on the agricultural, energy and transport sector. Thus, in reality the share of firms with a tax reduction is likely to be smaller than one.

7 Conclusion

In this paper, I empirically assess a tax credit program in Germany to estimate the causal impact of a capital cost reduction on firm input choices. To evaluate the success of this program, I go beyond a firm investment analysis and study the effect on employment of different labor types and spillover effects.

I find that firms increase both their investment and employment substantially and that this then translates to higher revenue. For employment, the effect occurs through hiring additional employees, including a sizable share that were unemployed. I do not find a shift in the employment composition by skill types. Nevertheless, industries with higher dependence on ICT technology are more likely to shift to high-skilled labor. Finally, spillovers further increase the employment effect.

These results are encouraging for the use of investment incentive programs in fiscal policies. The fact that there is a benefit for unemployed and low-skilled individuals is important from a welfare perspective since it is believed that they have more to gain.

However, more research is needed to assess whether investment incentives are the most efficient way of improving economic outcomes and how programs directly targeted to employees compare during the short-term and the long-term.

Whether the results will transfer to other settings is open for debate. Since tax credits reduce capital costs independent of firm profits they should be appropriate throughout the business cycle and effective in economic downturns. However, firms in the German tax credit program were particularly hard hit and had much more potential to catch up.

The result on the larger shift towards high-skilled among high share ICT industries should be a warning sign even though I do not find changes on average. Given that nowadays ICT has become more important in the production process and that robots already take on many simple tasks, incentivizing investments in the future may lead to less beneficial outcomes for unemployed and low-skilled individuals.

References

Abel, A. B. (1980). Empirical investment equations: An integrative framework. *Carnegie-Rochester Conference Series on Public Policy*, 12:39 – 91.

Abel, A. B. and Blanchard, O. J. (1986). The present value of profits and cyclical movements in investment. *Econometrica*, 54(2):249–273.

Abel, A. B. and Eberly, J. C. (1994). A unified model of investment under uncertainty. *The American Economic Review*, 84(5):1369–1384.

Acemoglu, D. and Autor, D. (2011). Chapter 12 - skills, tasks and technologies: Implications for employment and earnings. In Card, D. and Ashenfelter, O., editors, *Handbook of Labor Economics*, volume 4 of *Handbook of Labor Economics*, pages 1043 – 1171. Elsevier.

Akerman, A., Gaarder, I., and Mogstad, M. (2015). The Skill Complementarity of Broadband Internet. *The Quarterly Journal of Economics*, 130(4):1781–1824.

Axtell, R. L. (2001). Zipf distribution of u.s. firm sizes. *Science*, 293(5536):1818–1820.

Bade, F.-J. and Alm, B. (2010). Endbericht zum Gutachten Evaluierung der Gemeinschaftsaufgabe "Verbesserung der regionalen Wirtschaftsstruktur" (GRW) durch einzelbetriebliche Erfolgskontrolle für den Förderzeitraum 1999-2008 und Schaffung eines Systems für ein gleitendes Monitoring. Research report, Federal Ministry for Economic Affairs and Energy Germany.

Becker, S. O., Egger, P. H., and von Ehrlich, M. (2010). Going nuts: The effect of eu structural funds on regional performance. *Journal of Public Economics*, 94(9 - 10):578 – 590.

Becker, S. O., Egger, P. H., and von Ehrlich, M. (2013). Absorptive capacity and the growth and investment effects of regional transfers: A regression discontinuity design with heterogeneous treatment effects. *American Economic Journal: Economic Policy*, 5(4):29–77.

Blanchard, O. and Perotti, R. (2002). An empirical characterization of the dynamic effects of changes in government spending and taxes on output. *The Quarterly Journal of Economics*, 117(4):1329–1368.

Bond, S. and Reenen, J. V. (2007). Chapter 65 microeconometric models of investment and employment. *Handbook of Econometrics*, 6:4417 – 4498.

Bronzini, R. and de Blasio, G. (2006). Evaluating the impact of investment incentives: The case of italy's law 488/1992. *Journal of Urban Economics*, 60(2):327 – 349.

Busso, M., Gregory, J., and Kline, P. (2013). Assessing the incidence and efficiency of a prominent place based policy. *American Economic Review*, 103(2):897–947.

Cerqua, A. and Pellegrini, G. (2014). Do subsidies to private capital boost firms' growth? a multiple regression discontinuity design approach. *Journal of Public Economics*, 109:114 – 126.

Chirinko, R. S. (2000). Investment Tax Credits. Working Paper Series 243, CESifo.

Combes, P.-P. and Gobillon, L. (2015). Chapter 5 - the empirics of agglomeration economies. In Duranton, G., Henderson, J. V., and Strange, W. C., editors, *Handbook of Regional and Urban Economics*, volume 5 of *Handbook of Regional and Urban Economics*, pages 247 – 348. Elsevier.

Criscuolo, C., Martin, R., Overman, H. G., and Reenen, J. V. (2016). The Causal Effects of an Industrial Policy. Discussion Paper 1113, Centre for Economic Performance.

Cummins, J. G., Hassett, K. A., Hubbard, R. G., Hall, R. E., and Caballero, R. J. (1994). A reconsideration of investment behavior using tax reforms as natural experiments. *Brookings Papers on Economic Activity*, 1994(2):1–74.

Dettmann, E., Brachert, M., and Titze, M. (2016). Identifying the effects of place-based policies - causal evidence from germany. Working Paper 5901, CESifo.

Dustmann, C. and Glitz, A. (2015). How do industries and firms respond to changes in local labor supply? *Journal of Labor Economics*, 33(3):711–750.

Edgerton, J. (2010). Investment incentives and corporate tax asymmetries. *Journal of Public Economics*, 94(11):936 – 952.

Eichfelder, S. and Schneider, K. (2014). Tax incentives and business investment: Evidence from german bonus depreciation. Working Paper 4805, CESifo.

Fuest, C., Peichl, A., and Siegloch, S. (2018). Do higher corporate taxes reduce wages? micro evidence from germany. *American Economic Review*, 108(2):393–418.

Garicano, L., Lelarge, C., and Van Reenen, J. (2016). Firm size distortions and the productivity distribution: Evidence from france. *American Economic Review*, 106(11):3439–79.

Gathmann, C., Helm, I., and Schönberg, U. (2017). Spillover effects of mass layoffs. Working paper, mimeo.

Glaeser, E. L. and Gottlieb, J. D. (2008). The Economics of Place-Making Policies. *Brookings Papers on Economic Activity*, 39(1 (Spring):155–253.

Goolsbee, A. (1998). Investment tax incentives, prices, and the supply of capital goods*. *The Quarterly Journal of Economics*, 113(1):121–148.

Greenstone, M., Hornbeck, R., and Moretti, E. (2010). Identifying agglomeration spillovers: Evidence from winners and losers of large plant openings. *Journal of Political Economy*, 118(3):536–598.

Hall, R. E. and Jorgenson, D. W. (1967). Tax policy and investment behavior. *American Economic Review*, 57(3):391–414.

Hamermesh, D. S. (1993). *Labor Demand*. Princeton University Press Princton, N.J.

Hassett, K. A. and Hubbard, R. G. (2002). Chapter 20 - tax policy and business investment. In Auerbach, A. J. and Feldstein, M., editors, *Handbook of Public Economics*, volume 3 of *Handbook of Public Economics*, pages 1293 – 1343. Elsevier.

Hayashi, F. (1982). Tobin's marginal q and average q: A neoclassical interpretation. *Econometrica*, 50(1):213–224.

Heckman, J. J., Ichimura, H., and Todd, P. E. (1997). Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64(4):605–654.

Hethey-Maier, T. and Schmieder, J. F. (2013). Does the use of worker flows improve the analysis of establishment turnover? evidence from german administrative data. *Schmollers Jahrbuch*, 133(4):477–510.

House, C. L. and Shapiro, M. D. (2008). Temporary investment tax incentives: Theory with evidence from bonus depreciation. *American Economic Review*, 98(3):737–68.

Kline, P. and Moretti, E. (2014a). Local economic development, agglomeration economies, and the big push: 100 years of evidence from the tennessee valley authority. *The Quarterly Journal of Economics*, 129(1):275.

Kline, P. and Moretti, E. (2014b). People, places, and public policy: Some simple welfare economics of local economic development programs. *Annual Review of Economics*, 6(1):629–662.

Krusell, P., Ohanian, L. E., Ríos-Rull, J.-V., and Violante, G. L. (2000). Capital-skill complementarity and inequality: A macroeconomic analysis. *Econometrica*, 68(5):1029–1053.

Maffini, G., Xing, J., and Devereux, M. P. (2016). The impact of investment incentives: evidence from uk corporation tax returns. Working Paper 16/01, Oxford University Centre for Business Taxation.

Mertens, K. and Ravn, M. O. (2013). The dynamic effects of personal and corporate income tax changes in the united states. *American Economic Review*, 103(4):1212–47.

Moretti, E. (2010). Local multipliers. *American Economic Review*, 100(2):373–77.

Moretti, E. (2011). Chapter 14 - local labor markets. In Card, D. and Ashenfelter, O., editors, *Handbook of Labor Economics*, volume 4 of *Handbook of Labor Economics*, pages 1237 – 1313. Elsevier.

Moretti, E. and Thulin, P. (2013). Local multipliers and human capital in the united states and sweden. *Industrial and Corporate Change*, 22(1):339–362.

Neumark, D. and Simpson, H. (2015). Chapter 18 - place-based policies. In Duranton, G., Henderson, J. V., and Strange, W. C., editors, *Handbook of Regional and Urban Economics*, volume 5 of *Handbook of Regional and Urban Economics*, pages 1197 – 1287. Elsevier.

Romer, C. D. and Romer, D. H. (2010). The macroeconomic effects of tax changes: Estimates based on a new measure of fiscal shocks. *American Economic Review*, 100(3):763–801.

Sauer, S. and Strobel, T. (2015). Technologieintensität von investitionen nach branchen. Report 15, CESifo Institute.

Schmucker, A., Seth, S., Ludsteck, J., Eberle, J., and Ganzer, A. (2016). Establishment history panel 1975-2014. Fdz-datenreport, Institute for Employment Research.

Stierwald, A. and Wiemers, J. (2003). Auswirkungen der Gemeinschaftsaufgabe zur "Verbesserung der regionalen Wirtschaftsstruktur" auf die Investitionsfaehigkeit: Eine einzelwirtschaftliche Wirkungsanalyse fuer Betriebe des Verarbeitendes Gewerbes in Ostdeutschland. Discussion Paper 185, Halle Institute for Economic Research.

Suárez Serrato, J. C. and Zidar, O. (2016). Who benefits from state corporate tax cuts? a local labor markets approach with heterogeneous firms. *American Economic Review*, 106(9):2582–2624.

Summers, L. H. (1981). Taxation and corporate investment: A q-theory approach. *Brookings Papers on Economic Activity*, (1):67–140.

Yagan, D. (2015). Capital tax reform and the real economy: The effects of the 2003 dividend tax cut. *American Economic Review*, 105(12):3531–63.

Zwick, E. and Mahon, J. (2017). Tax policy and heterogeneous investment behavior. *American Economic Review*, 107(1):217–48.

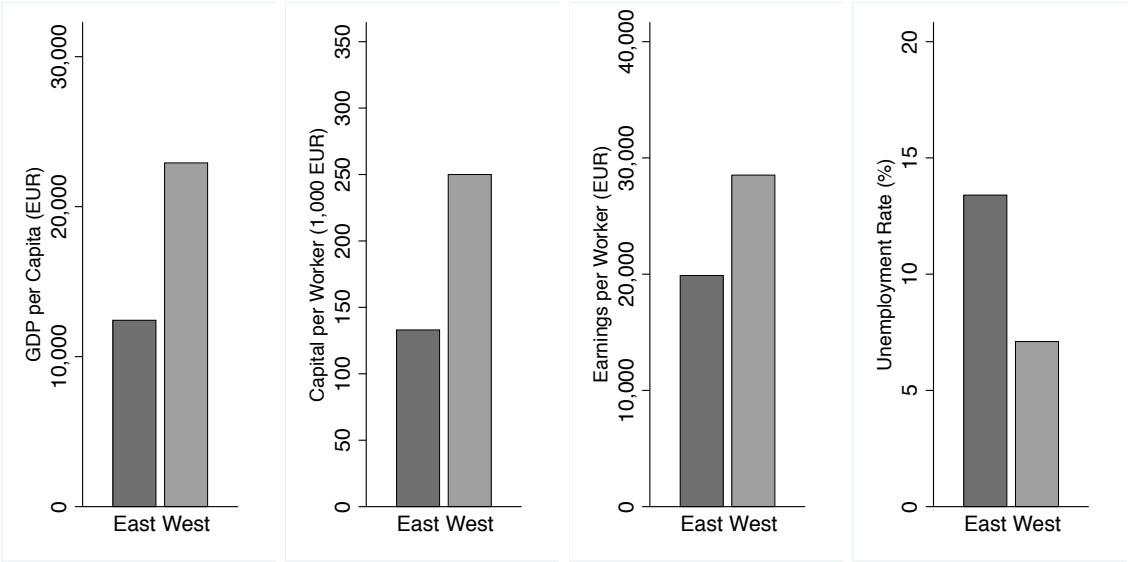


Figure 1: Economic Indicators for West and East Germany for 1991-1994

Note: The bars represent averages over the years 1991-1994. Data source for Panel (a)-(c) is the Federal Statistical Office. Data source for Panel (d) is the Federal Employment Institute. The unemployment rate is based on figures of June of each year. All of Berlin is counted towards the statistics of East Germany.

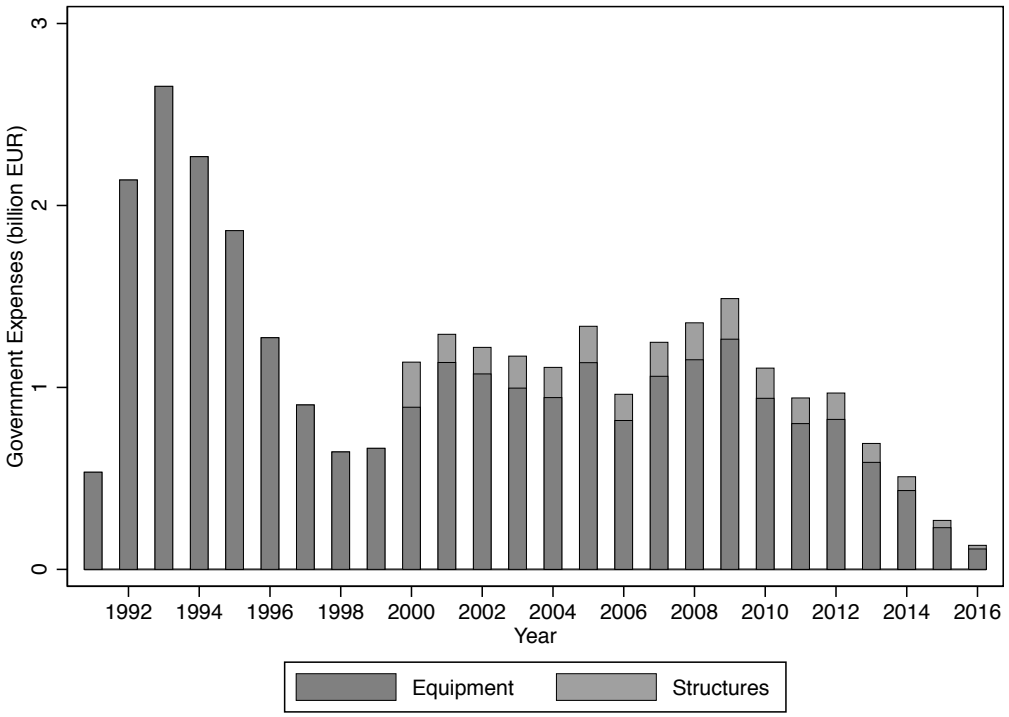


Figure 2: Government Expenses for Tax Credits by Year and Investment Type

Note: Information taken from subsidy reports of the German government. Values that are available only in Deutsche Mark were converted to euro using the official conversion rate. Figures on government expenses get revised over time due to additional information. The presented figures are taken from the most recent report available for each year respectively to reflect the most current information status. Expenses after 2016 excluded.

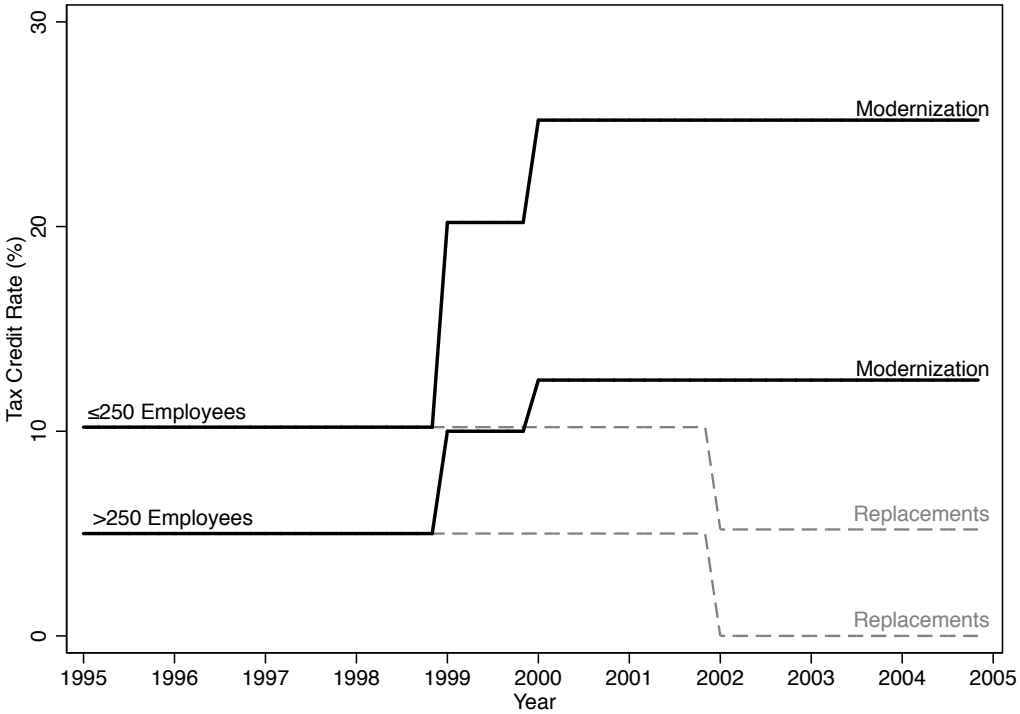


Figure 3: Equipment Tax Credit Rates for Manufacturing Firms

Note: The shown tax credit rates apply to manufacturing firms in most parts of East Germany excluding Berlin for equipment investment that start and end at the same day. For Berlin and in some years for areas close to Berlin rates were lower. There was a slight increase in the tax credit rate for modernization investments in regions close to the Polish or Czech border starting in 2001. Changes in tax credit rate were usually accompanied by phase-out periods to allow a constant rate for longer-lasting investment processes.

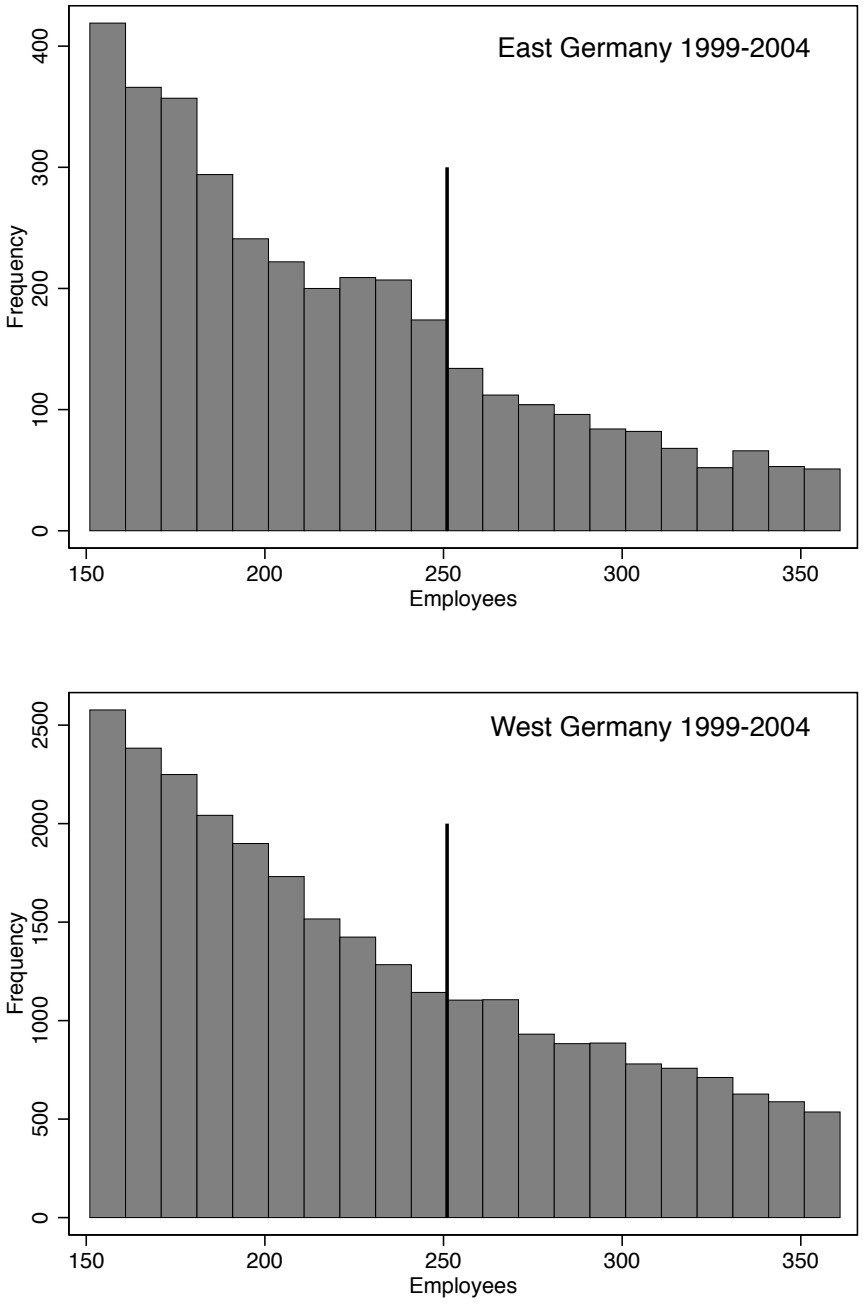


Figure 4: Firm Size Distribution of Manufacturing Industry

Note. Data from AFID establishment panel. The sample consist of all observations of manufacturing firms for East and West Germany excluding Berlin between 1999 and 2004. Firm size is based on total number of all employees but excludes vocational trainees.

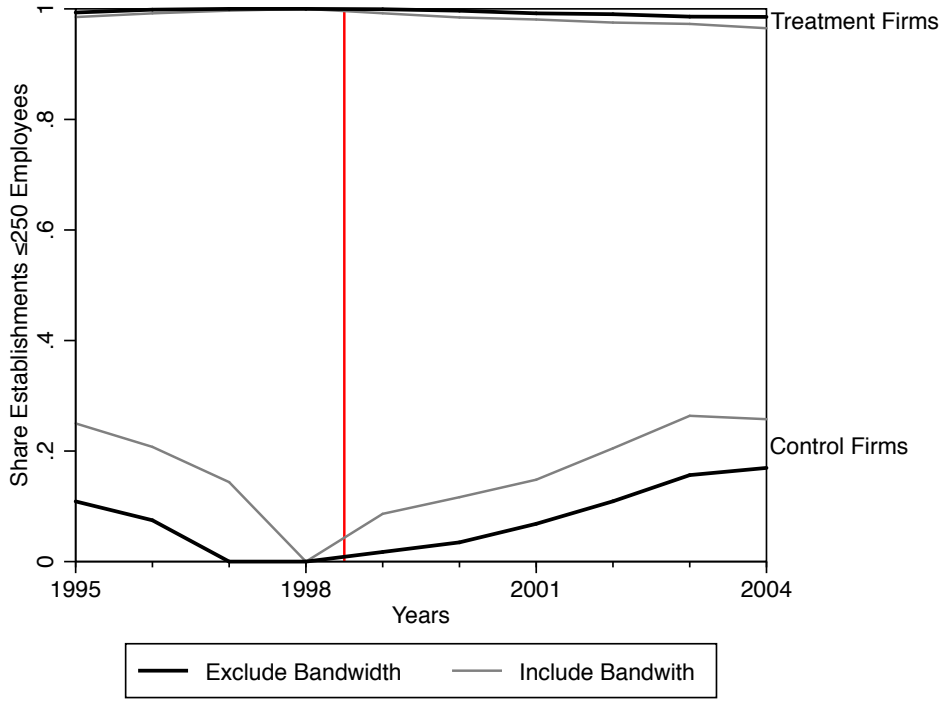


Figure 5: Movement of Treatment and Control Group Around Firm Size Cutoff

Note: Data from AFID establishment panel. The figure shows the share of firms in the treatment and control group below the firm size cutoff. Firm size is defined as total employee head count minus vocational trainees. The treatment group consists of firms with 250 or fewer employees. The sample selection for 'Exclude Bandwidth' is according to main text and specifies the exclusion of firms and observations around the firm size cutoff. The case of 'Include Bandwidth' follows the same sample selection, but keeps firms and observations around the size cutoff.

Table 1: Descriptive Statistics for Manufacturing Firms

Panel A. AFID Data

	West Germany	East Germany		
	All	All	Treatment	Control
No. of Employees (mean)	155.19	89.42	91.88	508.19
No. of Employees (median)	57	47	75	414
Investing (%)	86.0	86.0	94.0	97.0
Investments (thousand EUR)	1,267.50	1,172.35	805.96	8,605.11
Revenue (million EUR)	30.75	13.03	11.39	101.73
Part of Multi-Establishment Firm	22.0	22.0	9.98	15.76
Observations	357,662	64,154	16,374	1,266

Panel B. BHP Data

	West Germany	East Germany		
	All	All	Treatment	Control
No. of Employees (mean)	129.24	81.09	89.89	468.66
No. of Employees (median)	47	41	72	404.5
Full-time employees (mean)	111.63	70.95	80.56	424.02
Average Daily Wage Full-time	85.86	58.16	59.54	78.23
Share College Degree (%)	6.43	10.58	11.27	14.70
Share High-Skilled Occupation (%)	12.63	13.06	13.08	16.87
Share Vocational Trainees (%)	3.89	5.30	4.33	4.13
Observations	212,924	37,239	9,180	1,000

Note: Statistics are based on firms in the manufacturing sector for the years 1995-2004 excluding Berlin. The number of observations is based on the according statistic for number of employees. The group of all manufacturing firms includes those with more than 20 employees to allow for a better comparison between AFID and BHP data. Treatment and control group are according to the estimation sample.

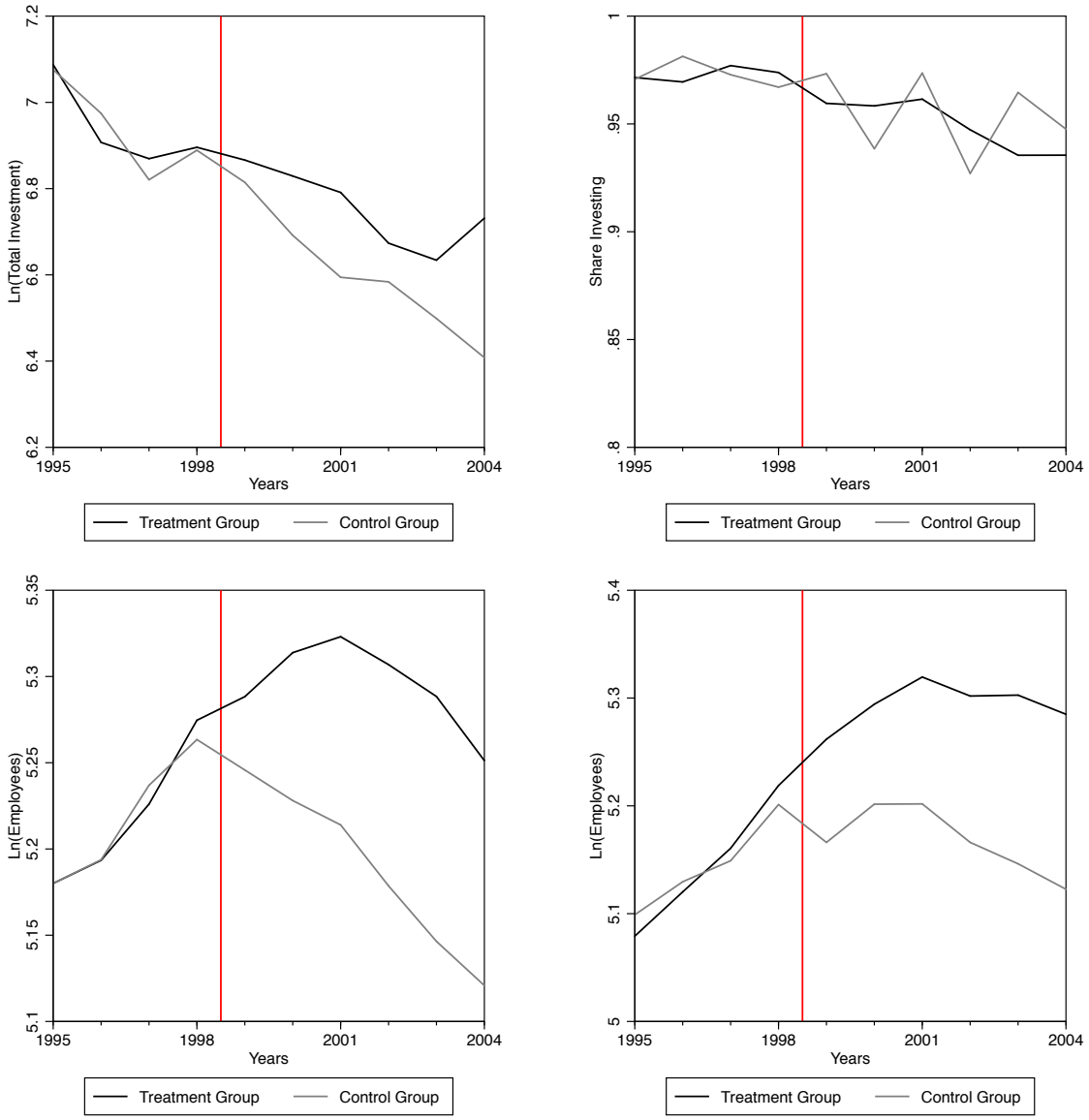


Figure 6: Average Firm Investment and Employment By Treatment Status

Note. The figures report raw means for each outcome by treatment and control group over the period 1995 to 2004. Outcome variables are the log of total investment in the upper left panel, share of firms with positive total investment in the upper right panel, log of employment based on AFID data in the lower left panel and log of employment based on BHP data in lower right panel. For each data point, I subtract the group mean for the pre-treatment period (1995-1998) and add the pooled mean for the same period for facilitating a comparison of trends.

Table 2: Difference-in-Differences - Investment

	Dependent Variable:				
	log (Total Investment)	log (Equipment Investment)	Investing (1/0)	$\frac{\text{Total Invest}_t}{\text{Avg Tot Invest}}$	$\frac{\text{Equip Invest}_t}{\text{Avg Tot Invest}}$
	(1)	(2)	(3)	(4)	(5)
Treated	0.234**	0.251***	-0.000	0.171**	0.167**
	(0.089)	(0.092)	(0.008)	(0.081)	(0.068)
Observations	15,275	15,071	15,900	15,865	15,865

Note. Data from AFID establishment panel. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of total investment in column (1), log of equipment investment in column (2), a dummy for having positive total investment in column (3), total investment over the average total investment between 1995-2004 in column (4) and equipment investment over the average total investment between 1995-2004 in column (5). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 3: Difference-in-Differences - Specification Robustness Investment

	(1)	(2)	(3)	(4)
	Dependent Variable: Log(Total Investment)			
Treated	0.234**	0.218**	0.201**	0.270***
	(0.089)	(0.090)	(0.090)	(0.094)
Observations	15,275	14,547	15,275	13,047
	Dependent Variable: Log(Equipment Investment)			
Treated	0.251***	0.231**	0.227**	0.285***
	(0.092)	(0.093)	(0.092)	(0.099)
Observations	15,071	14,347	15,071	12,880
	Dependent Variable: Investing (1/0)			
Treated	-0.000	-0.000	-0.004	-0.002
	(0.008)	(0.008)	(0.007)	(0.009)
Observations	15,900	15,148	15,900	13,555

Note. Data from AFID establishment panel. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of total investment in the upper panel and the log of equipment investment in the lower panel. Column (1) reproduces the coefficient from Table 2. Column (2) excludes firms with volatile investment measured by growth of total investment in 1997 above the 95th percentile, column (3) only includes log average firm wage, firm fixed effects and year fixed effects as control and column (4) is conditional on a a sample of single-establishment firms and excluding firms with volatile investment measured by growth of total investment in 1997 above the 95th percentile. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

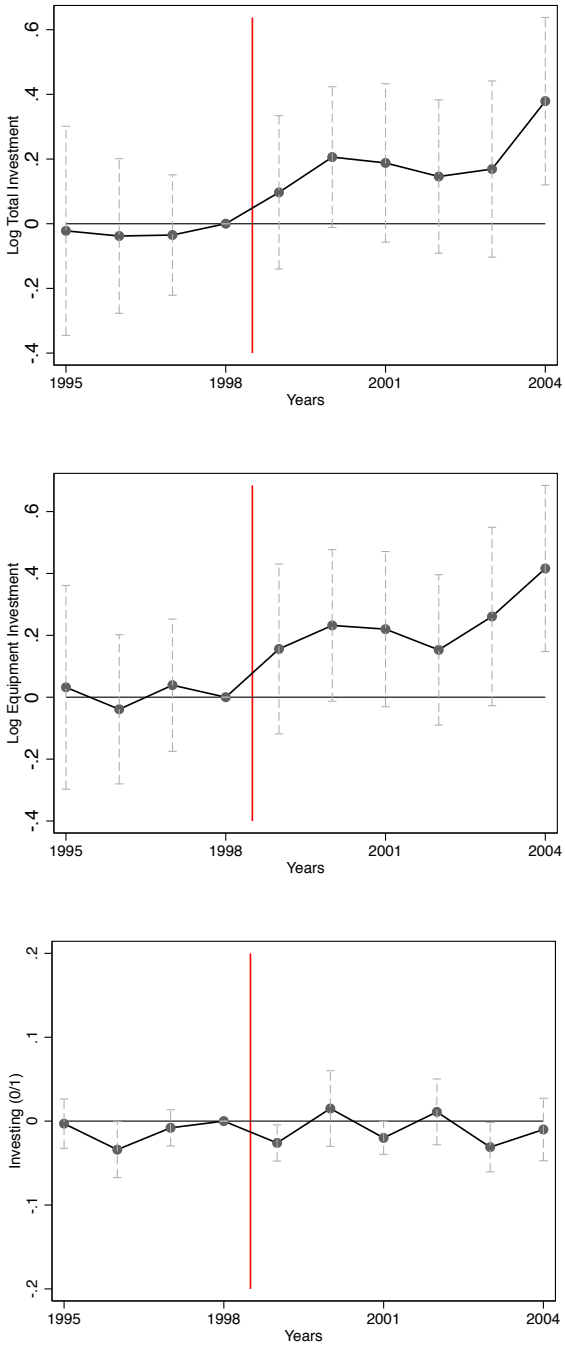


Figure 7: Dynamic Effect of Policy Change on Investment Behavior

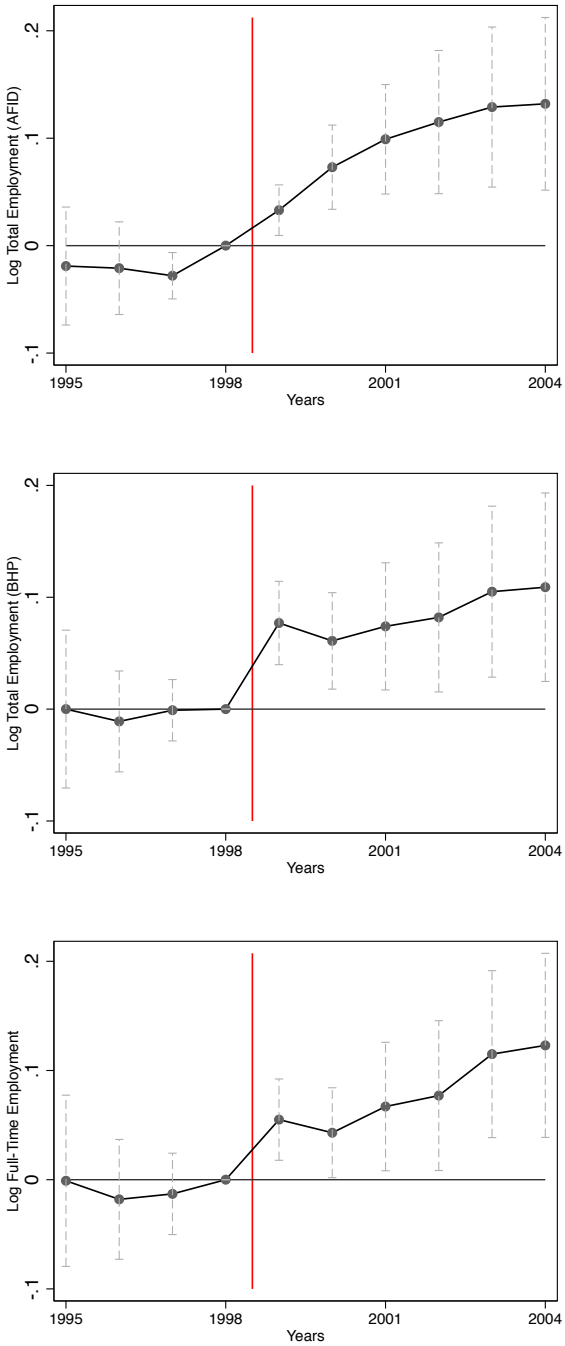
Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total investment in the upper panel, log of equipment investment in the middle panel and a dummy for having positive total investment in the lower panel. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Firms with volatile investment, measured by growth of total investment in 1997 above the 95th percentile, are excluded. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 4: Difference-in-Differences - Employment

	Dependent Variable: Log Employment of				
	Total (AFID)	Total (BHP)	Regular	Full-Time	Full-Time-Regular
	(1)	(2)	(3)	(4)	(5)
Treated	0.113***	0.087**	0.076**	0.088**	0.087**
	(0.031)	(0.037)	(0.036)	(0.038)	(0.038)
Observations	17,637	10,116	10,116	10,116	10,116

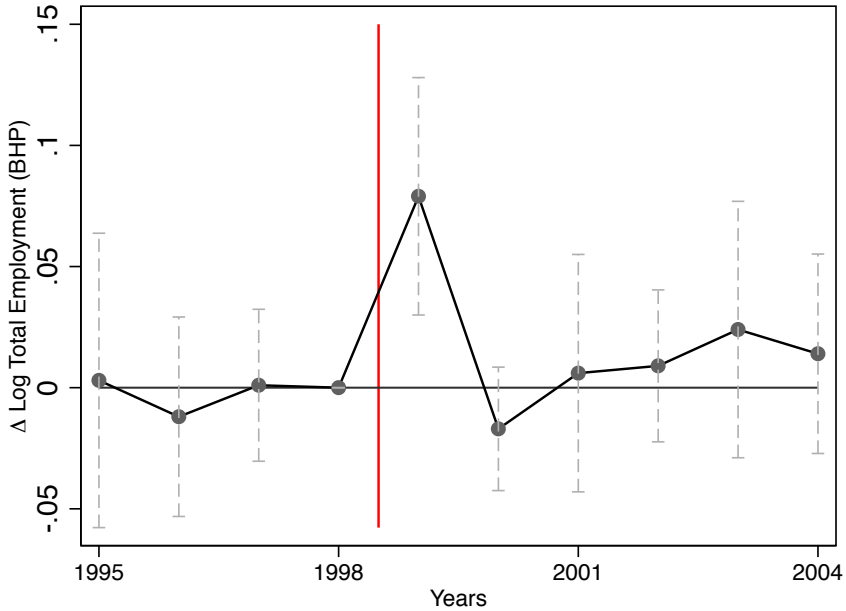
Note. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of total employment in column (1) and column (2), log of regular employees in column (3), log of full-time employees in column (4) and log of regular full-time employees in column (5). Data from AFID in column (1) and BHP in column (2)-(5). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Figure 8: Dynamic Effect of Policy Change on Firm Employment



Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total employment. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Figure 9: Dynamic Effect of Policy Change on Firm Employment Growth



Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total employment. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 5: Difference-in-Differences - Analysis of Flow Data

	Dependent Variable:					
	Flows All Employees			Flows Unemployment		
	Net	In	Out	Net	In	Out
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.086**	0.102**	0.022	0.031	0.050	0.018
	(0.037)	(0.041)	(0.032)	(0.019)	(0.034)	(0.026)
Observations	10,099	10,099	10,099	10,099	10,099	10,099

Note. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of cumulated net flows growth in column (1), the log of cumulated inflow growth in column (2), the log of cumulated outflow growth in column (3), the log of cumulated net flow growth from or to unemployment in column (4), the log of cumulated inflow growth from or to unemployment in column (5) and the log of cumulated outflow growth from or to unemployment in column (6). Growth rate is defined as a flow over past year total employment. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

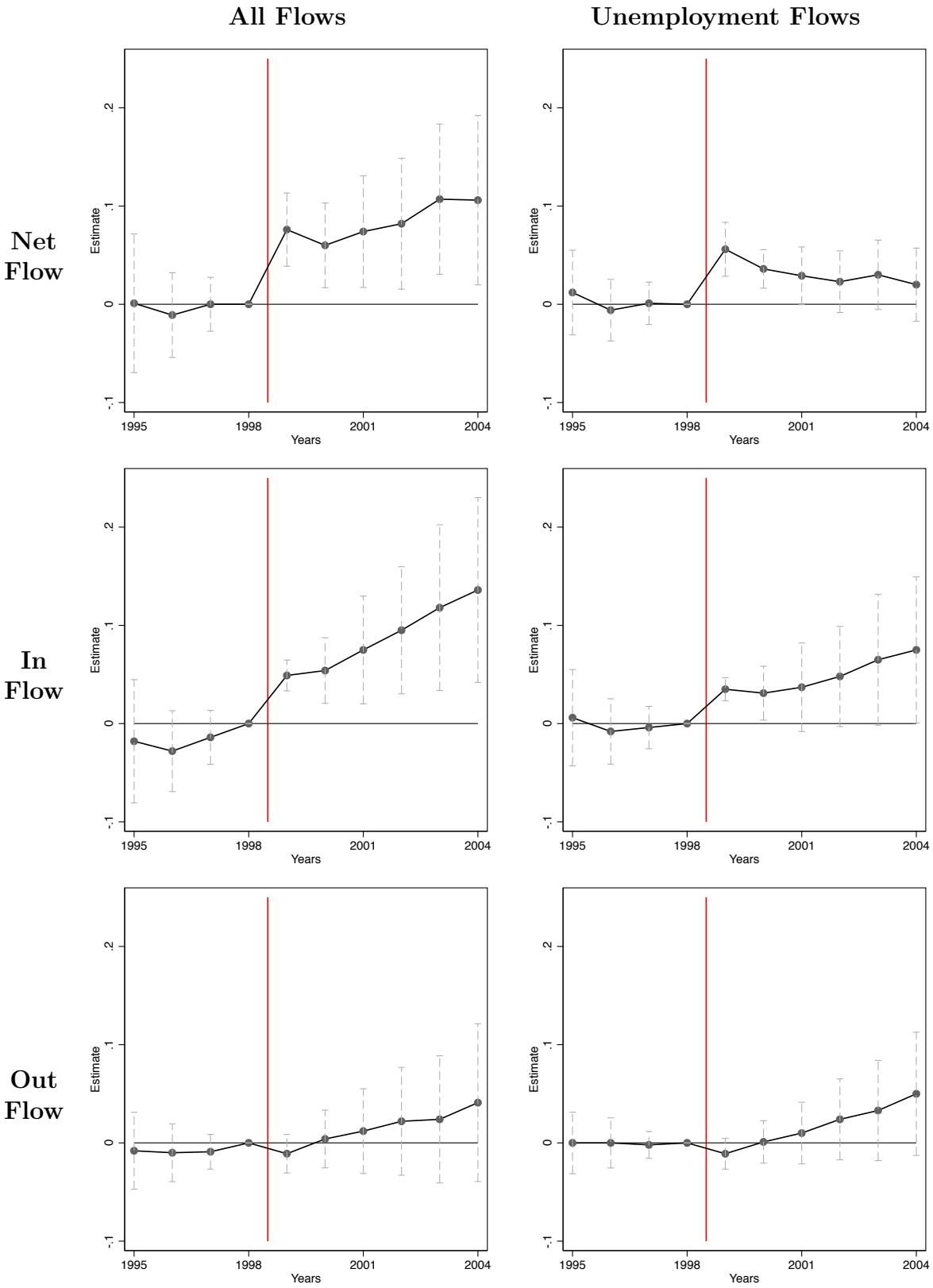


Figure 10: Dynamic Effect By Flow Type

Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of cumulated flow growth. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 6: Differences-in-Differences - Skill Ratios

	Dependent Variable: Log Ratio of				
	College Educated	Full-time College Educ	High-Skilled	Manual Labor	Qualified Within Manual Labor
	(1)	(2)	(3)	(4)	(5)
Treated	-0.011	0.014	-0.027	-0.019	0.004
	(0.032)	(0.031)	(0.043)	(0.045)	(0.043)
Average Share (%)	11.6	12.3	13.4	74.2	42.2
Obs	8,521	8,466	8,623	8,968	8,008

Note. Each coefficient is estimated from different regression following main specification (14). The dependent variables are the log of college educated vs. non-college educated in column (1), the log of full-time college educated vs. full-time non-college educated in column (2), the log of high-skilled occupations vs. low-skilled occupations in column (3), the log of manual occupations vs. service occupations in column (4) and the log of qualified manual occupations vs. unqualified manual occupations in column (5). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * $p<0.10$, ** $p<0.05$, *** $p<0.01$

Table 7: Differences-in-Differences - Employment Shares By Skill Level

	Dependent Variable: Log Ratio of				
	College Educated	Full-time College Educ	High-Skilled	Manual Labor	Qualified Within Manual Labor
	(1)	(2)	(3)	(4)	(5)
Treated	-0.021	0.005	-0.041	-0.009	-0.006
	(0.029)	(0.028)	(0.041)	(0.045)	(0.043)
Treated x ICT Share Investing IAB-LIAB (demeaned)	0.793**	0.710*	0.880**	-0.358	0.439
	(0.374)	(0.389)	(0.439)	(0.582)	(0.46)
Observations	8,521	8,466	8,623	8,968	8,008
Treated	-0.013	0.014	-0.032	-0.022	0.002
	(0.030)	(0.029)	(0.041)	(0.045)	(0.045)
Treated x ICT Investment Share ifo (demeaned)	0.745*	0.800*	1.081**	0.390	-0.322
	(0.435)	(0.418)	(0.426)	(0.682)	(0.720)
Observations	8,521	8,466	8,623	8,968	8,008
Treated	-0.015	0.011	-0.036	-0.012	-0.010
	(0.030)	(0.029)	(0.042)	(0.045)	(0.043)
Treated x ICT Capital Share EU KLEMS (demeaned)	1.984	1.775	2.877*	-1.084	1.591
	(1.371)	(1.420)	(1.677)	(2.201)	(2.082)
Observations	8,521	8,466	8,623	8,968	8,008
Treated	-0.001	0.024	-0.019	-0.030	0.015
	(0.033)	(0.031)	(0.044)	(0.044)	(0.048)
Treated x ICT Share CapEx U.S. Economic Census (demeaned)	1.378	1.113	1.514	-2.525	3.568*
	(1.388)	(1.276)	(1.378)	(1.988)	(2.142)
Observations	8,521	8,466	8,623	8,968	8,008

Note. Each coefficient is estimated from different regression following main specification (14) including interaction term. The dependent variables are the log of college educated vs. non-college educated in column (1), the log of full-time college educated vs. full-time non-college educated in column (2), the log of high-skilled occupations vs. low-skilled occupations in column (3), the log of manual occupations vs. service occupations in column (4) and the log of qualified manual occupations vs. unqualified manual occupations in column (5). The interaction term is the share of firms investing ICT in a given year based on the IAB-LIAB dataset in panel 1, the share of ICT investment based on Sauer, Strobel (2015) in panel 2, the ICT capital share for Germany based on EU KLEMS data and the ICT capital expenditure share based on U.S. Economic Census. Interaction terms are demeaned. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 8: Difference-in-Differences - Effect on Output

	Dependent Variable:				
	Log Revenue (1)	Log Domestic Revenue (2)	Log Domestic Manuf Revenue (3)	Log Revenue (low exporting) (4)	Log Products (5)
β	0.011	0.083**	0.080*	0.076	0.022
	(0.037)	(0.038)	(0.040)	(0.050)	(0.032)
Obs	15,906	15,898	15,897	11,689	15,547

Note. Each coefficient is estimated from different regression following Specification 14. The dependent variables are log of total revenue in column (1), log of domestic revenue in column (2), log of domestic revenue for manufacturing output in column (3), log of revenue in column (4) and log of the number of distinct products in column (5). The sample in column (4) is conditional on an exporting share of below 0.15. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 9: Impact of Excluded Bandwidth Around Cutoff

	Excluded Bandwidth:				
	None	[238,262]	[225,275]	[213,287]	[200,300]
	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: Log(Total Investment)				
Treated	0.209**	0.208**	0.218**	0.212**	0.210**
	(0.082)	(0.085)	(0.090)	(0.096)	(0.098)
Observations	15,559	15,014	14,547	14,168	13,862
	Dependent Variable: Log(Equipment Investment)				
Treated	0.207**	0.189**	0.231**	0.218**	0.215**
	(0.087)	(0.090)	(0.093)	(0.102)	(0.103)
Observations	15,354	14,813	14,347	13970	13,674
	Dependent Variable: Investing (0/1)				
Treated	-0.000	-0.004	-0.000	-0.002	-0.001
	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)
Observations	16,186	15,625	15,071	14,763	14,446
	Dependent Variable: Log(Employees) AFID				
Treated	0.091***	0.105***	0.113***	0.108***	0.109***
	(0.030)	(0.031)	(0.031)	(0.033)	(0.034)
Observations	18,377	18,007	17,637	17,287	16,997
	Dependent Variable: Log(Employees) BHP				
Treated	0.080**	0.080**	0.087**	0.107**	0.114**
	(0.036)	(0.036)	(0.037)	(0.044)	(0.046)
Observations	10,406	10,256	10,116	9,876	9,706
	Dependent Variable: Log(Domestic Revenue)				
Treated	0.058*	0.074*	0.083**	0.088**	0.089**
	(0.032)	(0.038)	(0.038)	(0.040)	(0.039)
Observations	16,988	16,382	15,898	15,510	15,191

Note. Data from AFID establishment panel and BHP. Each coefficient is estimated from different regression following main specification (14). The excluded firm and observations (for employment only firm) are according to the column titles. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 10: Impact of Firm Size Interval

	Firm Size Interval					
	[80,225], [275,750] (1)	[54,225], [275,1125] (2)	[45,225], [275,1350] (3)	[40,225], [275,1500] (4)	[34,225], [275,1800] (5)	[20,225], [275,3000] (6)
	Dependent Variable: Log(Total Investment)					
Treated	0.165*	0.244***	0.211**	0.218**	0.284***	0.327***
	(0.093)	(0.091)	(0.091)	(0.090)	(0.089)	(0.096)
Observations	6,307	10,459	12,821	14,547	17,043	23,945
	Dependent Variable: Log(Machinery Investment)					
Treated	0.163*	0.262***	0.224**	0.231**	0.273***	0.300***
	(0.092)	(0.093)	(0.092)	(0.093)	(0.092)	(0.095)
Observations	6,218	10,332	12,657	14,347	16,799	23,566
	Dependent Variable: Investing (0/1)					
Treated	0.008	0.013	0.003	-0.000	-0.003	-0.020***
	(0.010)	(0.010)	(0.008)	(0.008)	(0.008)	(0.007)
Observations	6,464	15,625	13,308	15,148	17,819	25,494
	Dependent Variable: Log(Employees) AFID					
Treated	0.091**	0.101***	0.105***	0.113***	0.135***	0.128***
	(0.039)	(0.033)	(0.032)	(0.031)	(0.034)	(0.033)
Observations	7,964	12,968	15,626	17,637	20,455	27,717
	Dependent Variable: Log(Employees) BHP					
Treated	0.097**	0.097**	0.089**	0.087**	0.101***	0.115***
	(0.045)	(0.040)	(0.037)	(0.037)	(0.035)	(0.034)
Observations	4,540	7,406	9,086	10,116	11,826	19,244
	Dependent Variable: Log(Domestic Revenue)					
Treated	0.090**	0.096**	0.082**	0.083**	0.111***	0.121***
	(0.045)	(0.041)	(0.040)	(0.038)	(0.041)	(0.042)
Observations	6,731	11363	13,987	15,898	18,701	26,796

Note. Data from AFID establishment panel and BHP. Each coefficient is estimated from different regression following main specification (14). The sample consists of firms according to the size intervals given in the column titles. Except for log employment these intervals apply to observations as well. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 11: Placebo Cutoffs Within Treatment Group

	Dependent Variable				
	Log Total Investment	Log Machinery Investment	Investing (1/0)	Log Employees (AFID)	Log Employees (BHP)
	(1)	(2)	(3)	(4)	(5)
Cutoff: 80					
Treated	-0.043	-0.038	-0.030*	0.033	0.006
	(0.078)	(0.076)	(0.016)	(0.026)	(0.030)
Observations	6,552	6,448	6,803	9,939	5,410
Cutoff: 100					
Treated	-0.027	0.023	-0.024**	0.039	0.009
	(0.068)	(0.072)	(0.012)	(0.025)	(0.027)
Observations	8,623	8,510	9,011	12,237	7,036
Cutoff: 125					
Treated	0.010	0.040	-0.020*	0.020	0.026
	(0.079)	(0.076)	(0.011)	(0.027)	(0.036)
Observations	10,385	10,247	10,886	14,025	7,916

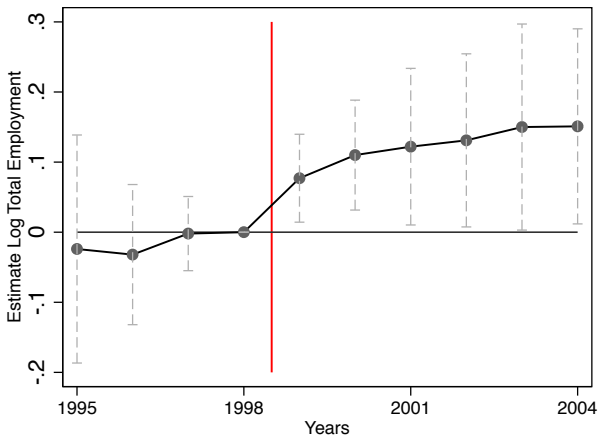
Note. Data from AFID establishment panel and BHP. Each coefficient is estimated from different regression following a modified version of specification (14). I change the firm size interval to between 50 and 250 employees. I set a cutoff for treatment and control group to 80 in the first panel, 100 in the second panel and 125 in the third panel. I exclude firms in an interval between -24 and +24 of the cutoff. Except for log employment these intervals apply to observations as well. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 12: Spillover Effects on Labor Inputs

	Dependent Variable: Log Total Employment		
	(1)	(2)	(3)
Firm size interval [40,225],[275,1500]			
Treated Firm	0.081**	0.088**	0.082**
	(0.036)	(0.036)	(0.036)
Regional Share Firms	0.272***	0.289***	0.253***
Below Cutoff	(0.095)	(0.090)	(0.084)
Observations	10,116	10,116	10,116
Firm size interval [30,225],[275,1500]			
Treated Firm	0.091**	0.097***	0.092***
	(0.035)	(0.035)	(0.035)
Regional Share Firms	0.188**	0.200**	0.171**
Below Cutoff	(0.091)	(0.090)	(0.077)
Observations	13,156	13,156	13,156
Firm size interval [20,225],[275,1500]			
Treated Firm	0.109***	0.114***	0.110***
	(0.037)	(0.038)	(0.037)
Regional Share Firms	0.148*	0.162*	0.138*
Below Cutoff	(0.087)	(0.084)	(0.072)
Observations	19,184	19,184	19,184
Firm size interval [10,225],[275,1500]			
Treated Firm	0.118***	0.119***	0.121***
	(0.039)	(0.038)	(0.037)
Regional Share Firms	0.054	0.058	0.041
Below Cutoff	(0.078)	(0.075)	(0.066)
Observations	34,019	34,019	34,019
Fixed Effects	Firm, Year	Firm, Industry-Year	Firm, Industry-Year, Area-Year

Note. Each coefficient is estimated from different regression following specification 16. The dependent variable is log total employment. Additional controls are log average firm wage and the fixed effects specified in each column. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Figure 11: Dynamic Effect of Spillovers on Labor Inputs



Note. Coefficients θ_p from estimation of dynamic specification (17). The dependent variable is the log of total employment. The firm size interval is [20,225],[275,1500]. Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and area (Bundesland)-year fixed effects. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.

Table 13: Spillover Effects on Labor Inputs

	Dependent Variable: Log Total Employment		
	(1)	(2)	(3)
Firm size interval [40,225],[275,1500]			
Treated Firm	0.081**	0.074*	0.088**
	(0.039)	(0.038)	(0.040)
Regional Share Same	0.308	0.250	0.170
Industry Below Cutoff	(0.188)	(0.181)	(0.207)
Regional Share Other	0.166*	0.173**	0.203**
Industry Below Cutoff	(0.089)	(0.085)	(0.081)
Observations	10,180	10,180	10,180
Firm size interval [30,225],[275,1500]			
Treated Firm	0.090**	0.084**	0.098**
	(0.037)	(0.037)	(0.039)
Regional Share Same	0.255	0.179	0.155
Industry Below Cutoff	(0.195)	(0.184)	(0.222)
Regional Share Other	0.110	0.119	0.125*
Industry Below Cutoff	(0.078)	(0.074)	(0.069)
Observations	13,240	13,240	13,240
Firm size interval [20,225],[275,1500]			
Treated Firm	0.113***	0.103***	0.122***
	(0.039)	(0.038)	(0.040)
Regional Share Same	0.207	0.123	0.189
Industry BelowCutoff	(0.151)	(0.136)	(0.177)
Regional Share Other	0.106	0.117*	0.110*
Industry Below Cutoff	(0.073)	(0.066)	(0.062)
Observations	19,328	19,328	19,328
Firm size interval [10,225],[275,1500]			
Treated Firm	0.122***	0.107***	0.129***
	(0.039)	(0.038)	(0.039)
Regional Share Same	0.237*	0.169	0.244*
Industry Below Cutoff	(0.130)	(0.111)	(0.136)
Regional Share Other	0.014	0.019	0.011
Industry Below Cutoff	(0.066)	(0.059)	(0.052)
Observations	34,334	34,334	34,334
Fixed Effects	Firm, Year	Firm, Year, Region trends	Firm, Industry-Year, Area-Year

Note. Each coefficient is estimated from different regression following specification 16. The dependent variable is log total employment. Additional controls are firm-level pre-treatment wage growth trends and the fixed effects specified in each column. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Table 14: Spillover Effects to Service Industry

	Dependent Variable: Log Total Employment		
	(1)	(2)	(3)
Firm size interval [40,225],[275,1500]			
Regional Share Manuf Firms	0.002	-0.017	0.008
Below Cutoff	(0.040)	(0.036)	(0.033)
Observations	37,313	37,313	37,313
Firm size interval [30,225],[275,1500]			
Regional Share Manuf Firms	-0.016	-0.026	-0.000
Below Cutoff	(0.035)	(0.032)	(0.031)
Observations	50,390	50,390	50,390
Firm size interval [20,225],[275,1500]			
Regional Share Manuf Firms	-0.029	-0.033	-0.002
Below Cutoff	(0.031)	(0.027)	(0.023)
Observations	76,003	76,003	19,184
Firm size interval [10,225],[275,1500]			
Regional Share Manuf Firms	-0.026	-0.034	0.022
Below Cutoff	(0.032)	(0.028)	(0.023)
Observations	142,836	142,836	142,836
Fixed Effects	Firm, Year	Firm, Area-Year	Firm, Industry-Year, Area-Year

Note. Each coefficient is estimated from different regression following specification 16 excluding the interaction term of the firm size cutoff on a sample of firms in the service industry. The dependent variable is log total employment. Additional controls are firm-level pre-treatment wage growth trends and the fixed effects specified in each column. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

Appendices

A Competing Programs

The main text interprets the results of the empirical analysis as the effect of the investment tax credit program described in section 1. During the period of analysis, further support was available to firms in East Germany through other programs. There might be the concern, that these programs confound the analysis. I describe the two most important competing programs and the differences to the tax credit program. I discuss how relying simultaneously on cross-sectional and time series variation can control for the changes in the other programs. I also provide additional robustness tests that suggest little influence if any from the competing programs. Taken together, the evidence supports the claim that the results in the main text are caused by the tax credit program.

A.1 Special Depreciation Allowances

The German government introduced special depreciation allowances to all firms in East Germany in 1991.[39] In general, depreciations distribute the purchasing costs of fixed assets over the life span of the asset to match them to their generated profits. As an expense they reduce the amount of taxes on profits. The policy allowed the depreciation of 50% of an asset's value directly after purchase. The remaining value followed standard depreciation practices by depreciating equal amounts each year over the life span of the asset. This change did not influence the total amount of depreciation for the asset, however it moved the resulting tax benefit to earlier years. This reduces capital costs since temporal discounting makes a current benefit more valuable than an equally sized future one.

Given that there was only a shift in timing, the seemingly large special depreciation allowance converted to a smaller capital cost reduction. House and Shapiro (2008) calculate the capital cost reduction for a 50% special depreciation allowance of a similar program in the U.S. They find that for short-lived assets of 5 years, there is a reduction in capital costs by 1.26%. For long-lived assets of 20 years, the capital cost reduction is close to 5%. Equipment typically has a life span of 10 years and therefore the allowances provided relatively small benefits.

The special depreciation program lasted until the end of 1998 and thus coincided with the change in the tax credit rates. However, the percentage of special depreciation allowances was the same for all firms and did not induce any differential treatment be-

[39] The program is very similar to those studied in House and Shapiro (2008), Maffini et al. (2016) and Zwick and Mahon (2017). Eichfelder and Schneider (2014) analyze the German program comparing firm in East Germany with those in West Germany.

tween firms below and above the tax credit rate cutoff. Since I include time fixed effects in my main specification, I control for the discontinuation of the special depreciation allowances.[40]

Nevertheless, firms from different industries likely differ in their capital structure, particularly in the life span of their fixed assets. This leads to a different capital cost reduction from special depreciation allowances, with industries with longer-lived assets receiving a larger reduction. If the capital structure of industries correlates with firm size, special depreciation can influence treatment along the same dimensions as the tax credit program. However, since this type of influence is likely industry-specific, the use of industry-year fixed effects can control for this type of influence.[41]

The simultaneity of the changes in both tax policies was not a coincidence. The special depreciation allowances were stopped with the goal of incorporating a similar capital cost reduction through tax credits. This is the reason for the increase in the tax credit rate for all firms in 1999. At the same time, the policy shifted the focus towards small firms, introducing the additional differential treatment along firm size. There is no evidence that the focus on small firms was due to different pre-policy trends or anticipated differential shocks by firm size. The focus on small firms seems to rather reflect a general preference for small firms, which is common for policies worldwide. Arguments brought forward are the large share of small firms in an economy and unfair competition advantages of large firms.

A.2 German Joint Task Program

The German Joint Task Program was another program through which firms in East Germany could receive a reduction in investment costs. The program already existed before reunification of Germany and is currently the main place-based policy instrument in Germany.[42] After reunification, the main focus was put on regions in East Germany. The program allocated funds for infrastructure investments and firm-specific investments to each eligible federal state in Germany. Based on a general set of rules, each state then decided on the specific criteria for the selection and support of investment projects. Since funds were limited, decisions on allocation was based on a competitive basis. Important criteria were the expected profitability of firms and the commitment to keep or hire

[40] I consider the differential percentage change in capital costs from the equal reduction in special depreciation allowances due to a differentially distorted capital cost between firms below and above the tax credit rate cutoff as negligible.

[41] Any remaining interactions between the two programs should not hamper the qualitative results assuming that special depreciation allowances were less important since both programs were pure investment cost shocks. However, since the exact size of the shock would be unclear, it would be difficult to state the effects in terms of an elasticity.

[42] Dettmann et al. (2016) analyze the program using regional variation in West Germany and applying RDD techniques. Stierwald and Wiemers (2003) and Bade and Alm (2010) study the program using firm-level variation.

employees. Firms had to apply to these grants before initiating their investment project and were uncertain about the success of their application. Thus, this program mixed a reduction in capital costs with a selection on firm characteristics and specific employment decisions with uncertainty in the success and the value of a given grant.

Overall, during the years in question the joint task program was not as important as tax credits. Using information from the IAB-establishment survey for 1997, 76% of firms state that they used investment tax credits, whereas 34% used GRW grants. Most firms with GRW grants used tax credits at the same time as well. The value of funds distributed to East German manufacturing firms were roughly half of the tax credits. The program also focused on large investment projects with a priority for the opening of new firms.

Similar to the investment tax credits, the program provided larger incentives for smaller firms. However, the differential treatment stayed stable over time, with constant maximum subsidy rates for small and large firms respectively and little evidence for changes in applying this maximum subsidy rate. Thus, comparing firms below and above the tax credit rate cutoff before and after the policy change should identify the direct investment grant independent of the joint task program. For a few regions, maximum rates were adjusted at the end of 1996 and at the end of 1999, but the dynamic results in the main text suggest no effect for years before 1999 and a direct response to the investment tax credit change in 1999.

Table A.1: Difference-in-Differences - Effect on Investment (AFID)

	Dependent Variable:		
	$\frac{\text{Total Invest}_t}{\text{Avg Tot Invest}}$	$\frac{\text{Equip Invest}_t}{\text{Avg Tot Invest}}$	$\frac{\text{Structure Invest}_t}{\text{Avg Tot Invest}}$
	(1)	(2)	(3)
Treatment	0.167**	0.174**	-0.007
	(0.080)	(0.067)	(0.031)
Observations	15,865	15,865	15,865

Note. Data from AFID establishment panel. Each coefficient is estimated from different regression following main specification (14). The dependent variables are total investment over the average total investment between 1995-2004 in column (1), equipment investment over the average total investment between 1995-2004 in column (2) and investment in structures over the average total investment between 1995-2004 in column (3). Additional controls are log average firm wage, firm fixed effects, industry-year fixed effects and labor market-year fixed effects. Standard errors in parentheses are clustered at the regional level: * p<0.10, ** p<0.05, *** p<0.01

To further verify my claims, I consider a robustness test. In contrast to the tax credits, the GRW grants had preferential rates for small firms for investments in structures. Table A.1 compares the estimation result for investments in structures and equipment investments for manufacturing firms in East Germany. For better comparability, I focus on the investment measure in the main text that combines extensive and intensive margin effects. The first two columns reproduce the results for scaled total investment, with an estimate of 0.167 and scaled equipment investment, with an estimate of 0.174. In

comparison, the effect for scaled investments in structures of -0.007 is economically and statistically insignificant. The difference in the effect for equipment and structures is in line with the regulations of the investment tax credit program but would be surprising for the GRW program.

B Theoretical Framework

In this section I provide detailed steps for solving the firm production model. I leave out general explanations that are covered in the main text.

In the model there are many heterogeneous firms i in many regions r with each firm producing one differentiated good. The production function for each firm is

$$F(K_i, U_i, S_i) = Y_i = A_i A_{ir} \left[(a_K K_i^{\rho} + a_S S_i^{\rho})^{\frac{\mu}{\rho}} + a_U U_i^{\mu} \right]^{\frac{1}{\mu}}, \tag{18}$$

where output Y_i is produced using capital K_i, low-skilled labor U_i and high-skilled labor S_i as input factors. The nested CES-production function is governed by the parameters a_K, a_S and a_U. The elasticity of substitution between low-skilled workers and capital (or high-skilled workers) is $\frac{1}{1-\mu}$ and the elasticity of substitution between high-skilled and capital is $\frac{1}{1-\rho}$. Production also depends on a firm-specific production parameter A_i and a region-specific productivity shifter A_{ir}.

Each firm maximizes their profit facing the downward sloping inverse demand curve

$$p_i = B Y_i^{-\frac{1}{\eta^D}}, \tag{19}$$

and has a capital rental rate r, wage to low-skilled labor w_U and wage to high-skilled labor w_S. Each firm receives a tax credit τ_i on their capital.

The firm maximization problem then is

$$\max_{K_i, S_i, U_i} p_i Y_i - (1-\tau_i) r K_i - w_S S_i - w_U U_i, \tag{20}$$

subject to equations (18) and (19).

I plug in equations (18) and (19) to obtain an unconstrained maximization problem. The resulting first order conditions from the maximization problem are

$$(1-\tau_i) r = \left(1 - \frac{1}{\eta^D}\right) B a_K Y_i^{1-\mu-\frac{1}{\eta^D}} X^{\frac{\mu-\rho}{\rho}} K_i^{\rho-1} (A_i A_{ir})^{\mu} \tag{21}$$

$$w_S = \left(1 - \frac{1}{\eta^D}\right) B a_S Y_i^{1-\mu-\frac{1}{\eta^D}} X^{\frac{\mu-\rho}{\rho}} S_i^{\rho-1} (A_i A_{ir})^{\mu} \tag{22}$$

$$w_U = \left(1 - \frac{1}{\eta^D}\right) B a_U Y_i^{1-\mu-\frac{1}{\eta^D}} U_i^{\mu-1} (A_i A_{ir})^{\mu}, \tag{23}$$

where

$$X_i = a_K K_i^{\rho} + a_S S_i^{\rho}, \tag{24}$$

and where I substitute parts of the expression with the production function.

The FOCs serve to study how firms adjust their production process after a change in

tax credit rate. Considering that tax credits are a form of reduction in capital costs, it is straightforward to show that

$$d\ln c_K = \frac{d(1-\tau_i)r}{(1-\tau_i)r} = -\frac{rd\tau_i}{(1-\tau_i)r} = \frac{d\tau_i}{1-\tau_i}, \tag{25}$$

where c_K are capital costs of a firm.

With that in mind, the total derivative of equation (21) written as elasticity with respect to the tax credit rate is

$$-1 = \left(1-\mu-\frac{1}{\eta^D}\right)\frac{dY_i}{d\tau_i}\frac{1-\tau_i}{Y_i} + \frac{\mu-\rho}{\rho}\frac{dX_i}{d\tau_i}\frac{1-\tau_i}{X_i} + (\rho-1)\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} + \mu\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}. \tag{26}$$

To simplify the calculation of the total derivative based on equation (22), first I divide left- and right-hand side with equation (21). The resulting elasticity is

$$\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i} = \frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} - \frac{1}{1-\rho}. \tag{27}$$

The total derivative of equation (23) written as elasticity is

$$\frac{dU_i}{d\tau_i}\frac{1-\tau_i}{U_i} = \left(1-\frac{1}{(1-\mu)\eta^D}\right)\frac{dY_i}{d\tau_i}\frac{1-\tau_i}{Y_i} + \frac{\mu}{1-\mu}\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}. \tag{28}$$

The equations (26)–(28) depend on the elasticity of Y_i and X_i with respect to the tax credit rate. I totally differentiate equation (18) and (24) to receive

$$\frac{dY_i}{d\tau_i}\frac{1-\tau_i}{Y_i} = \frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} + \left(\frac{A_iA_{ir}}{Y_i}\right)^{\mu}\left(a_U U_i^{\mu}\frac{dU_i}{d\tau_i}\frac{1-\tau_i}{U_i} + a_K X_i^{\frac{\mu-\rho}{\rho}} K_i^{\rho}\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} + a_S X_i^{\frac{\mu-\rho}{\rho}} S_i^{\rho}\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i}\right) \tag{29}$$

$$\frac{dX_i}{d\tau_i}\frac{1-\tau_i}{X_i} = \rho a_K\frac{K_i^{\rho}}{X_i}\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} + \rho a_S\frac{S_i^{\rho}}{X_i}\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i}. \tag{30}$$

With equations (26)–(30), there are six unknown elasticities and five equations. For now I keep the elasticity of regional productivity and solve the system of equations for the rest of the elasticities. I do so by plugging in equation (27) into equation (30) and subsequently equation (30) into equation (26). I also plug equations (27) and (28) in equation (29). This leads to two equations and by subtracting one from the other, the elasticities for output and capital are

$$\frac{dY_i}{d\tau_i}\frac{1-\tau_i}{Y_i} = \eta^D\left(\frac{A_iA_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}}\left(1-\frac{a_S S_i^{\rho}}{X_i}\right) + \eta^D\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \tag{31}$$

$$\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} = \eta^D \left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + \frac{1}{1-\mu}\left(1 - \left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i}\right) - \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right)\frac{a_S S_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}. \quad (32)$$

Following from that, the elasticities for low-skilled and high-skilled labor are

$$\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i} = \left(\eta^D - \frac{1}{1-\mu}\right)\left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right)\frac{a_K K_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \quad (33)$$

$$\frac{dU_i}{d\tau_i}\frac{1-\tau_i}{U_i} = \left(\eta^D - \frac{1}{1-\mu}\right)\left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} + (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}. \quad (34)$$

To facilitate the interpretation of these elasticities, I substitute parts of the equations with the relationships given in the FOCs. I rearrange equations (21)–(23) as

$$\left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_K K_i^{\rho}}{X_i} = \frac{\eta^D}{\eta^D - 1}\frac{(1-\tau_i)r}{p_i}\frac{K_i}{Y_i} = Z s_{K_i} \quad (35)$$

$$\left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} X_i^{\frac{\mu}{\rho}} \frac{a_S S_i^{\rho}}{X_i} = \frac{\eta^D}{\eta^D - 1}\frac{w_S}{p_i}\frac{S_i}{Y_i} = Z s_{S_i} \quad (36)$$

$$\left(\frac{A_i A_{ir}}{Y_i}\right)^{\mu} a_U U_i^{\mu} = \frac{\eta^D}{\eta^D - 1}\frac{w_U}{p_i}\frac{U_i}{Y_i} = Z s_{U_i}, \quad (37)$$

where $Z = \frac{\eta^D}{\eta^D - 1}$, $s_{K_i} = \frac{(1-\tau_i)r}{p_i}\frac{K_i}{Y_i}$, $s_{S_i} = \frac{w_S}{p_i}\frac{S_i}{Y_i}$ and $s_{U_i} = \frac{w_U}{p_i}\frac{U_i}{Y_i}$, omitting the index i on the shares for .

With these terms the elasticities can be written as

$$\frac{dY_i}{d\tau_i}\frac{1-\tau_i}{Y_i} = \eta^D s_{K_i} Z + \eta^D \frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \quad (38)$$

$$\frac{dK_i}{d\tau_i}\frac{1-\tau_i}{K_i} = \left[\eta^D s_{K_i} + \frac{1}{1-\mu} s_{U_i} + \frac{1}{1-\rho} s_{S_i} - \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right) s_{U_i}\frac{a_S S_i^{\rho}}{X_i}\right] Z + (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \quad (39)$$

$$\frac{dS_i}{d\tau_i}\frac{1-\tau_i}{S_i} = \left[\left(\eta^D - \frac{1}{1-\rho}\right) s_{K_i} + \left(\frac{1}{1-\mu} - \frac{1}{1-\rho}\right) s_{U_i}\frac{a_K K_i^{\rho}}{X_i}\right] Z + (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \quad (40)$$

$$\frac{dU_i}{d\tau_i}\frac{1-\tau_i}{U_i} = \left(\eta^D - \frac{1}{1-\mu}\right) s_{K_i} Z + (\eta^D - 1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}. \quad (41)$$

To consider changes in the composition of the labor force by skill type, I use the ratio between high-skilled and low-skilled. We have

$$\frac{S_i}{U_i} = \left(\frac{w_s}{w_u}\frac{a_u}{a_s}\right)^{\frac{1}{\rho-1}} X_i^{\frac{\rho-\mu}{\rho(\rho-1)}} U_i^{\frac{\rho-\mu}{1-\rho}} \quad (42)$$

The elasticity with respect to the tax credit can be calculated using the previous elasticities as

$$\frac{d\frac{S_i}{U_i}}{d\tau_i}\frac{1-\tau_i}{\frac{S_i}{U_i}} = \frac{\mu-\rho}{(1-\mu)(1-\rho)}\frac{a_K K_i^\rho}{X_i} \tag{43}$$

The above results summarize the effect of a change in tax credit rate on firms that experience the rate change. Before I define A_{ir} more specifically, I first consider the effect of a change in tax credit rate on firms that do not experience the rate change.

In this case, taking into account that τ_j is not affected by changes in τ_i, the total derivatives from the FOCs are

$$0 = \left(1-\mu-\frac{1}{\eta^D}\right)\frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j} + \frac{\mu-\rho}{\rho}\frac{dX_j}{d\tau_i}\frac{1-\tau_i}{X_j} + (\rho-1)\frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} + \mu\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \tag{44}$$

$$\frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j} = \frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} - \frac{1}{1-\rho} \tag{45}$$

$$\frac{dU_j}{d\tau_i}\frac{1-\tau_i}{U_j} = \left(1-\frac{1}{(1-\mu)\eta^D}\right)\frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j} + \frac{\mu}{1-\mu}\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} \tag{46}$$

$$\frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j} = \frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}} + \left(\frac{A_j A_{ir}}{Y_j}\right)^\mu \left(a_U U_j^\mu \frac{dU_j}{d\tau_i}\frac{1-\tau_i}{U_j} + a_K X_j^{\frac{\mu-\rho}{\rho}} K_j^\rho \frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} + a_S X_j^{\frac{\mu-\rho}{\rho}} S_j^\rho \frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j}\right) \tag{47}$$

$$\frac{dX_j}{d\tau_i}\frac{1-\tau_i}{X_j} = \rho a_K \frac{K_j^\rho}{X_j}\frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} + \rho a_S \frac{S_j^\rho}{X_j}\frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j}. \tag{48}$$

I solve again for the five elasticities as before. I plug equations (45) and (48) into equation (44), and I use this result together with equation (45) and (46) in equation (47).

The elasticities are

$$\frac{dK_j}{d\tau_i}\frac{1-\tau_i}{K_j} = \frac{dS_j}{d\tau_i}\frac{1-\tau_i}{S_j} = \frac{dU_j}{d\tau_i}\frac{1-\tau_i}{U_j} = (\eta^D-1)\frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}, \tag{49}$$

and

$$\frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j} = \eta^D \frac{dA_{ir}}{d\tau_i}\frac{1-\tau_i}{A_{ir}}. \tag{50}$$

Focusing now on the indirect effect through changes in the regional productivity shifter, I first define the productivity shifter as

$$A_{ir} = \sum_{j\in S_r} Y_j^{\lambda_{ij}}, \tag{51}$$

where productivity for firm i in region r depends on the accumulated output of each firm i in region r weighted by the agglomeration elasticity λ_{ij}.

To simplify exposition, I assume that $\lambda_{ij} = \lambda$ and thus the productivity shifter is the same for all firms in a particular region, $A_r = A_{ir}$.

With this dependence, it is possible to provide a closed form solution for the elasticity

of regional productivity with respect to the tax credit rate τ_i:

$$\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \frac{d\sum Y_j^\lambda}{d\tau_i}\frac{1-\tau_i}{\sum Y_j^\lambda} = \sum\left(\lambda Y_j^{\lambda-1}\frac{dY_j}{d\tau_i}\right)\frac{1-\tau_i}{\sum Y_j^\lambda} = \frac{\lambda}{\sum Y_j^\lambda}\sum Y_j^\lambda \frac{dY_j}{d\tau_i}\frac{1-\tau_i}{Y_j}. \quad (52)$$

By plugging in equations (31) and (50), the elasticity of regional productivity to a change in tax credits of firm i is

$$\frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r} = \frac{\lambda\eta^D}{1-\lambda\eta^D}\frac{s_{K_i}Y_i^\lambda}{\sum Y_j^\lambda}Z. \quad (53)$$

For the case that several firms receive a change in the tax credit rate, it is possible to consider the total derivative

$$dA_r = \sum_i \frac{dA_r}{d\tau_i}d\tau_i = \sum_i \frac{dA_r}{d\tau_i}\frac{1-\tau_i}{A_r}\frac{A_r}{1-\tau_i}d\tau_i \quad (54)$$

With the result in equation (53), this transforms to

$$dA_r = \sum_i \frac{\lambda\eta^D}{1-\lambda\eta^D}\frac{s_{K_i}Y_i^\lambda}{\sum Y_j^\lambda}Z\frac{A_r}{1-\tau_i}d\tau_i = \frac{\lambda\eta^D}{1-\lambda\eta^D}Z\sum_i s_{K_i}Y_i^\lambda\frac{d\tau_i}{1-\tau_i} \quad (55)$$

To get to a meaningful expression in terms of elasticity, I make simplifying assumptions. I assume that the change in tax credit rate applies to a subset of all firms and is the same for this subset. Thus, $d\tau_i = d\tau$ for some firms and for the rest $d\tau_i = 0$. For the subset of firms experiencing change, I also assume that $\tau_i = \tau$. With that it is possible to rearrange terms to get

$$\frac{dA_r}{d\tau}\frac{1-\tau}{A_r} = \frac{\lambda\eta^D}{1-\lambda\eta^D}Z\sum_{i|d\tau_i\neq 0}\frac{s_{K_i}Y_i^\lambda}{\sum Y_j^\lambda} \quad (56)$$

C Data

This section complements the data section in the main text. I provide descriptive statistics for the complete set of variables used in the analysis. Further, I provide details on creating the final datasets, imputations of variables and specific sample selection. The following information pertains to the AFID Establishment-Panel[43], the Panel of Cost Structure[44] and the Establishment Panel Schmucker et al. (2016).

C.1 Descriptive Statistics

[43]Source: RDC of the Federal Statistical Office and Statistical Offices of the Länder, AFiD Establishment-Panel, 1995–2004, own calculations.

[44]Source: RDC of the Federal Statistical Office and Statistical Offices of the Länder, Panel of Cost Structure, 1999–2001, own calculations.

Table C.1: Descriptive Statistics BHP-Panel for Regression Sample

	Mean	P10	Median	P90	Count
Employment variables					
No. of total employees	127.10	42	78	253	10,180
No. of full-time employees	114.33	37	70	225	10,180
No. of regular employees	118.69	39	73	239	10,180
No. of regular full-time employees	114.29	37	70	225	10,180
log (total employees)	4.51	3.74	4.36	5.53	10,180
log (full-time employees)	4.39	3.61	4.25	5.42	10,180
log (regular employees)	4.44	3.66	4.29	5.48	10,180
log (regular full-time employees)	4.39	3.61	4.25	5.42	10,180
Δ log(total employees)	0.03	-0.11	.02	0.18	9,060
log (flow measure net all)	0.35	-0.31	0.19	1.06	10,163
log (flow measure hires all)	0.98	0.16	0.75	1.94	10,163
log (flow measure separations all)	0.61	0.14	0.53	1.20	10,163
log (flow measure net unemployment)	0.15	-0.22	0.06	0.58	10,163
log (flow measure hires unemployment)	0.55	0.09	0.43	1.11	10,163
log (flow measure separations unemployment)	0.40	0.08	0.35	0.76	10,163
Shares and ratios					
Share college (%)	11.68	1.52	8.70	25.00	9,106
Share college among full-time (%)	12.35	1.59	9.30	26.74	9,106
Share high-skilled occup (%)	13.42	2.00	10.20	26.98	9,106
Share manual occup among low-/mid-skilled (%)	74.06	41.48	80.65	88.24	9,105
Share mid-skilled occup among manual (%)	42.23	4.44	34.27	95.00	9,015
log (college / non-college)	-2.31	-3.77	-2.27	-1.06	8,560
log (college full-time / non-college full-time)	-2.23	-3.66	-2.20	-0.96	8,505
log (high-skilled occup / other occup)	-2.15	-3.53	-2.10	-0.97	8,666
log (manual occup / service occup)	1.30	-0.22	1.44	2.58	8,997
log (mid-skilled manual / low-skilled manual)	-0.53	-2.66	-0.69	1.97	8,030
Other firm variables					
Average daily wage full-time	61.64	40.68	59.05	85.12	9,106
log (average daily wage full-time)	4.08	3.71	4.08	4.44	9,106
Policy-relevant firm size	121.52	40	75	243	10,180
Labor market level variables					
Share employees in firm with $\leq$250 employees (%)	79.53	–	82.25	–	55
Share employees in firms of specific industry with $\leq$250 employees (%)	6.13	0.74	4.38	14.24	707
Share employees in firms of other industries with $\leq$250 employees (%)	73.41	51.69	75.95	92.90	707

Note: Statistics are based on firms in the manufacturing sector for the years 1995-2004 excluding Berlin. The descriptives of employment variables, and shares and ratios are additional based on the associated regression sample in the main text. The descriptives of Other firm variables are additional based on regression sample of employment. Labor market level and labor market-industry level variables are based on all manufacturing firms.

C.2 Prediction of Apprentices in AFID Establishment-Panel

For determining the firm size in context of the policy all employees count as one person independent of their employment contract or their working hours. The only exception are apprentices that have special contracts and therefore are not employees in legal terms. In the AFID establishment panel there is no information about apprentices. In order to have a better measure of the firm size, I use information from the Panel of Cost Structure (KSE). In this survey, firms were asked about their number of apprentices for the years 1999–2001 for a subsample of manufacturing firms. I create the share of apprentices within this dataset and then link this information to the AFID establishment panel with their firm identifier. For observations in which information on the share of apprentices is matched, I use the matched information. For observations that are part of a firm with matched apprentices information, I assume the share to be equal to the average share of matched information. I use information on the total number of employees in the AFID establishment panel in each year to impute the number of apprentices for these cases. For firms that have no matched apprentices information, I calculate the average share within 3-digit industry classifiers and assume it to be constant for all observations with missing information. For industries that have no single firm with matched apprentice information, I calculate the average share in the whole sample and assume it to be constant for all observations with missing information.

C.3 Predicting Marginally Employed Workers in the BHP

On the other hand, the firm size definition included marginally employed workers. The BHP records information about the number of marginally employed only since 1999 though. For classifying firms into small firms with at most 250 employees and large firms with more than 250 employees, it is important to impute marginally employed for prior years.

I use the following approach. First, I calculate the share of marginally employed for each establishment averaging over the years 1999 and 2000. For establishments that existed in these years, I assume that this share is constant over time and impute the number of marginally employed for years before 1999 directly. For establishments that did not exist in 1999 or afterwards, this direct method does not work. For these cases, I run a simple prediction model. I include all establishments in 1999 and 2000 with more than 20 total employees. I run the model separately for East and West Germany and include 3-digit industry classifiers and firm size. I allow the coefficient for firm size to vary within each industry. Since the prediction model is linear, the predicted share can be negative. In these cases, I set the share to zero. I also winsorize at the 99th-percentile to exclude extreme values in the prediction. I then proceed as before and assume a constant

share within each establishment.

Table C.2 provides information for the actual share, the calculated share and the predicted share.

C.4 Entry and Exit of Firms in the BHP

Since the difference-in-difference design makes use of the panel component, it is important to understand how firms enter and exit into the dataset. The BHP aggregates social security information on an establishment level and draws a 50% random sample of establishment identifiers. As noted in the data report by Schmucker et al. (2016), establishment identifiers can change when ownership changes or the establishment is merged with or split into different companies. Thus, some changes in the establishment identifier do not lead to changes in economic activity. To overcome this shortcoming, I merge an extension file provided by the IAB based on the methodology of Hethey and Schmieder (2010, 2013). In this file, entries and exits are classified according to flow information of coworkers from one establishment to another.[45] Essentially, if most employees move together from one establishment during its last year and constitute a large share of a newly created establishment, it is likely a change in the identifier without change in the economic structure. In contrast, if most workers move into unemployment or to various other establishments, it is likely an exiting firm. Table C.3 provides information from the extension file for the overall dataset and the specific sample in the data analysis. ID changes do not happen that often. In the regression sample 0.59% of entering establishments and 0.6% exiting establishments are affected.

The dataset provides information on the preceding / succeeding establishment identifier for entering / exiting establishments with identifier change. I use this information to match these establishments. I take into account that the establishment identifier can change more than once over the observation period. In case the information from identifier changes of entering and exiting establishments differs I choose the largest component. With this algorithm, I end up with 60 newly merged establishments in the regression sample which is a share of 3.24% of all establishments. There is also information on merges and split ups. However, these changes are more likely to influence production and I therefore consider them as distinct.

There is a second dimension to entry and exit that to the best of my knowledge has not been addressed in the official data description. It is concerned with the question whether an entry and exit into the dataset coincides with actual start and end of economic production. On various occasions, an establishment starts to exist with one or few employees and only in the next or a later year there is a jump to a distinctively larger number of employees. The same pattern in reverse happens before the exit of various establishments.

[45] This methodology uses matched employer-employee data that is not accessible to me.

Table C.2: Marginally employed in BHP Data

Regression Sample	Mean	P10	P50	P90	Count
Actual Shares 1999 - 2004	0.045	0.000	0.013	0.098	9,271
Actual Shares 1999 - 2000	0.040	0.000	0.012	0.086	3,380
Imputed within establishment 1995-1998	0.040	0.000	0.012	0.083	6,516
Imputed with prediction model 1995-1998	0.052	0.000	0.051	0.106	231
Absolute Difference Actual Value vs Imputed within Establishment 2001-2004	0.027	0.000	0.009	0.055	5,890
Absolute Difference Actual Value vs Prediction Model 2001-2004	0.057	0.004	0.037	0.102	5,891

Note: Statistics are based on establishments in the manufacturing sector for the years 1995-2004 excluding Berlin. The number of observations is based on the statistic for number of employees.

Table C.3: Descriptive Statistics - IAB-BHP Data

ID Changes Entering Establishments	
Overall (%)	0.17
Regression sample (%)	0.35
Regression sample (count)	56
ID Changes Exiting Establishments	
Overall (%)	0.16
Regression sample (%)	0.48
Regression sample (count)	77
Matching Regression Sample	
Affected establishments (%)	8.23
Affected establishments (count)	148
Matched establishments (%)	5.92
Matched establishments (count)	109

Note:

Table C.4: Descriptive Statistics

	Mean	Min	P10	P50	P 90	Max	Obs
Yearly employment growth rate	0.24	-1	-0.72	0.01	1.54	368	15,400
Yearly employment growth rate AFID	0.02	/	-0.11	0.01	0.15	/	15,867
Employment before increase by 200%		1	1	9	96	96	89
Employment after decrease by 67%		0	0	3	341	1,337	180
Share inflow from other establishments normal growth	0.39	0	0.0	0.38	0.71	1	14,802
Share outflow to other establishments normal growth	0.34	0	0.0	0.33	0.67	1	14,849
Share inflow from other establishments high growth	0.72	0	0.36	0.80	0.96	1	124
Share outflow to other establishments low growth	0.67	0	0.22	0.77	0.95	1	170

Table C.4 summarizes the incident of these patterns. Looking at the percentiles of yearly employment growth, the median firm has zero growth. At the 10th and 90th percentile, growth rates are already large with a decrease by 72% and an increase by 154%. The smallest and largest growth rates are extreme with either a fall to zero employees or an increase of 36,800%. This is not the case for the AFID data. Interestingly, many of these changes coincide with movement of employees from or to other (not necessarily the same) employers. The share of inflowing employees that worked before is 72% during growth of more than 200% but only 39% during lower growth excluding marginally employed. The share of outflowing employees that work afterwards is 67% during growth of less than -67% but only 34% during higher growth. This could potentially point to additional ID changes.

Because of these patterns, I assume that periods before growth of 200% or higher and periods after growth of -67% or lower are periods of non-production and I exclude

them as observations.[46] These cutoffs are guided by establishment behavior in the AFID dataset. To provide an understanding how different cutoffs affect the empirical analysis, Table C.5 provides estimates for employment using various cutoffs. Figure 12 shows the dynamic profile.

Table C.5: Static Estimation - Difference of Growth Rate Cutoff of Outliers

	1.5	2.0	2.5	3.0	5.0	10.0	none
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent Variable: Log(Number Employees)						
β	0.102***	0.076**	0.082**	0.080**	0.096**	0.063	0.125
	(0.029)	(0.030)	(0.035)	(0.035)	(0.046)	(0.052)	(0.089)
Obs	7,900	9,550	9,920	10,070	10,370	10,450	11,192
	Dependent Variable: Log(Ratio High-Skilled to Low-Skilled Occupation)						
β	-0.012	0.001	-0.005	-0.011	-0.020	-0.017	-0.030
	(0.033)	(0.031)	(0.033)	(0.032)	(0.029)	(0.030)	(0.031)
Obs	6,776	8,099	8,374	8,478	8,678	8,731	9,285

Note. Each coefficient is estimated from different regression following specification 14. The dependent variables are machinery investment over lagged revenue in column (1), machinery investment over lagged imputed capital in column(2), the natural logarithm of machinery investment in column (3) and a dummy for positive machinery investment in column (4). The upper panel shows estimates for regressions including firms with 40 to 1,500 employees in 1998, the lower one for firms with 80 to 750 employees. Further, the sample consist of single-establishment manufacturing firms in East Germany excluding Berlin with observations in 1995 and 2004. For all regressions, firms with imputed negative capital are excluded. In Column (1) and (2), values are truncated at the largest 5%. Standard errors in parentheses are clustered at the firm level: * p<0.10, ** p<0.05, *** p<0.01

[46] Although I consider it unlikely, it is possible, that the changes in employment happen because of large genuine adjustments in economic production from one year to another.

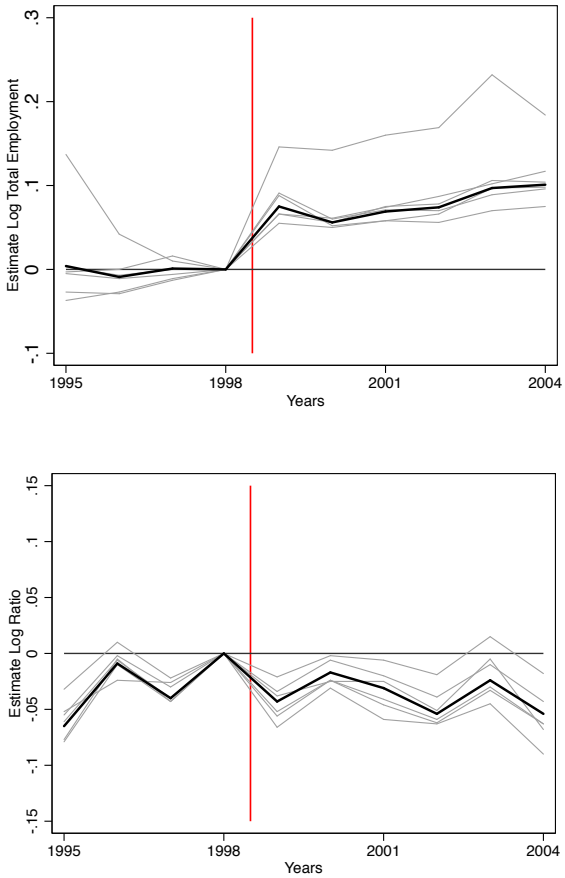


Figure 12: Dynamic Estimation - Difference of Growth Rate Cutoff of Outliers

Note. Coefficients from estimation of dynamic specification (15). The dependent variables are the log of total investment in the upper panel, log of equipment investment in the middle panel and a dummy for having positive total investment in the lower panel. Additional controls are log average establishment wage, establishment fixed effects, industry-year fixed effects and labor market-year fixed effects. Establishments with volatile investment, measured by growth of total investment in 1997 above the 95th percentile, are excluded. 95% confidence intervals are displayed for each coefficient using clustered standard errors at the regional level.